

EYE CARE MOBILE BASED APPLICATION

Oshani Tharanga Wickramasingha

IT13012502

Bachelor of Science Special (Honors) Degree in IT

Department of Information Technology

Sri Lankan Institute of Information Technology Sri Lanka

November 2016

EYE CARE MOBILE BASED APPLICATION

Wickramasingha Mudiyanseelage Oshani Tharanga

IT13012502

Dissertation submitted in partial fulfillment of the requirements for the B.Sc. Special
Honors Degree in IT

Department of Information Technology

Sri Lankan Institute of Information Technology Sri Lanka

November 2016

Declaration

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

Signature:

Date:

The above candidate has carried out research for the B.Sc. Special (Hons) degree in IT Dissertation under my supervision.

Signature of the Supervisor:

Date:

Abstract

At present people are suffering from eye disorders and sometimes they do not take care about their health. People are reluctant to wait in the queues because of their busy schedules as well as most of the time people do not have enough money to channel a doctor. "Eye Care" application can be introduced as an eye illness detecting application where it is mainly focusing on four eye illnesses namely Cataract, Conjunctivitis, Blepharitis and Chalazion. The uniqueness of this research is that the illnesses are identifying with a combination of two main techniques called image processing and knowledge base. In order to detect eye illnesses "Eye Care" uses Image Processing techniques and results are forwarding to the Knowledge Base. Then user needs to answer some questions given by the application and it detects the eye illness with some home precautions that user can take in order to prevent from the illness. "Eye Care" application's accuracy level is about 90%. There is no any commercially available mobile application which recognizes the above four eye illnesses. Main advantage of this application is that any user can use the application without taking help from others since the application is more user friendly when comparing to the existing applications that are available in the market. "Eye Care" application mainly targeting eye patients but it can be used by anyone who use android phones. Therefore this proposed system will be very useful for the people in order to check whether he/she is suffering from the above eye illnesses.

Key Words: Image Processing, Knowledge Base, Pre-Processing, Segmentation

Acknowledgment

We would like to sincerely thank each and every individual who guided and assists us on the process of completing our project. Our first honorable gratitude goes to Sri Lanka Institute of Information Technology, for providing the opportunity to expose our skills and abilities via performing a research project.

Furthermore we would like to acknowledge with much appreciation the crucial role of our project Ms Umanga Pilapitiya. She has tremendously guided and inspired our energies in the course of this research. She has also managed to remain in good contact with us in spite of the communication odds and occasional frustrations obtainable in our domestic front. She also provided a considerable number of the literature used in the analysis of this work. Frankly, we could not have made any significant head-way without the erudite guidance and assistance of our project supervisor.

We would also like to thank our lecturer in charge of “Comprehensive Design and Analysis Project” Mr. Jayantha Amararachchi for giving us necessary information and guidelines for us on how to successfully start, continue and complete our project by giving us accurate and clear instructions ensuring the successful completion of our project. His guidance and concern provided throughout the research of our project is thoroughly appreciated. And also I would like to thank Director Dr. I. L. M. Rifaz of National Eye Hospital for giving us details about eye illnesses and allow us to take photos of patients in order to launch our project.

Finally I convey my real sense of gratitude and thankfulness to all my friends and family members for their unconditional support and encouragement during my project work without which I would hardly be capable of producing this huge work.

Table of Contents

Declaration	i
Abstract	ii
Acknowledgment	iii
List of Tables.....	x
List of Figures	xi
List of Abbreviations.....	xii
1 INTRODUCTION	1
1.1 Background Context.....	1
1.1.1 Publications of facts by World Health Organization	1
1.1.2 Eye illness explanation.....	1
1.1.3 Image processing techniques.....	2
1.1.4 Current status of existing research	2
1.1.5 Module description.....	3
1.1.5.1 Image preprocessing and segmentation.....	3
1.1.5.2 Data extraction and classification	3
1.1.5.3 Creating rules of knowledge base	3
1.1.5.4 Data analyzing and finalize output.....	4
1.1.6 Topic significance	4
1.1.7 Categorization of eye applications	4
1.1.8 Existing eye applications.....	5
1.1.8.1 Eye Test app.....	5
1.1.8.2 Eye Care Plus app.....	6
1.1.8.3 Vision Test 2.0 app	6
1.1.8.4 EyeXam app	7
1.1.8.5 Visual Acuity Test app	8

1.1.8.6	White Eye Detector	8
1.1.8.7	Eye Exercises - Eye Care Plus.....	9
1.1.8.8	Amsler grid.....	9
1.1.8.9	Color Blind test.....	10
1.1.8.10	Pink Eye App	11
1.1.9	Researches based on image processing.....	11
1.1.10	Comparison of existing applications	12
1.1.11	Preprocessing and segmentation	13
1.1.11.1	Image preprocessing.....	13
1.1.11.2	Image segmentation.....	13
1.1.11.3	Histogram equalization	14
1.1.11.4	Detecting face, eyes, iris from the image.....	15
1.2	Research Gap.....	15
1.2.1	Gap reduction	16
1.3	Research Problem.....	16
1.4	Research Objectives	18
1.4.1	General objective	18
1.4.2	Specific objectives	18
1.5	Report Overview	19
1.6	Report Organization	20
2.	METHODOLOGY	22
2.1	Methodology	22
2.1.1	Referred models	22
2.1.2	Choice of Eye Care	22
2.1.3	Background reading	23
2.1.3.1	Gathering requirements	24
2.1.4	System design	25
2.1.4.1	Work breakdown structure.....	26
2.1.5	Overall description	27

2.1.5.1	Use case scenarios.....	31
2.2	Implementation and Testing.....	33
2.2.1	Implementation	33
2.2.2	User interfaces	36
2.2.2.1	Home screen	36
2.2.2.2	Option selection.....	37
2.2.2.3	About Eye Care	38
2.2.2.4	Uploading face	39
2.2.2.5	How to use Eye Care	40
2.2.2.6	Photo capturing.....	41
2.2.2.7	Camera opening.....	42
2.2.2.8	Distance calibration	43
2.2.3	Testing	44
2.2.3.1	Unit testing	44
2.2.3.2	System testing.....	44
2.2.3.3	Test cases	44
2.3	Research Findings	55
2.3.1	Image processing technique issues.....	55
2.3.2	Image processing module findings.....	55
2.3.2.1	Configuration of android application.....	55
2.3.2.2	Accessing smart phone memory.....	56
2.3.2.3	Exchanging images	56
2.3.2.4	Distance calibration.....	56
2.3.2.5	Measuring the light source	56
2.3.2.6	Detection of iris	57
3	RESULTS AND DISCUSSION	58
3.1	Results	58
3.1.1	Acceptance testing importance	58
3.1.2	Image preprocessing module outcomes	59

3.1.2.1	Image selection from album	59
3.1.2.2	Uploading image	60
3.1.2.3	Eyes detection and cropping.....	61
3.1.2.4	Iris detection and cropping	62
3.1.2.5	Results output	63
3.1.2.6	Eyes detection in self-camera.....	64
3.2	Discussion	65
3.2.1	Limitations	68
3.2.1.1	Limitations of iris detection and iris cropping	68
3.2.1.2	Quality of the image	69
3.2.1.3	Energy of light.....	69
3.2.2	Reasons to lower the accuracy of iris detection and cropping.....	69
3.3	Standards	70
3.3.1	Coding standards.....	70
3.3.1.1	Beginning Comments	70
3.3.1.2	Left and Right braces.....	71
3.3.1.3	Wrapping lines	72
3.3.1.4	White Spaces	72
3.3.1.5	Implementation comments.....	74
3.3.1.6	Standards of statements.....	75
3.3.1.7	Naming Conventions	78
3.3.1.8	Standards for Methods.....	79
3.3.1.9	Naming Convention standards	81
3.3.1.10	Standards for classes.....	82
3.3.2	UI standards	83
3.3.2.1	Form elements	83
3.3.2.2	Title	83
3.3.2.3	Description	83
3.3.2.4	Buttons.....	85

3.3.2.5	Interface text	86
3.3.2.6	UI-elements.....	87
3.3.2.7	Wording	88
3.3.2.8	Capitalization.....	89
3.3.2.9	Banners and background theme	90
3.4	Nonfunctional requirements.....	91
3.4.1	Reliability.....	91
3.4.2	Availability.....	91
3.4.3	Security	92
3.4.4	Maintainability	92
3.5	Other Requirements.....	93
3.5.1	User-friendliness	93
3.5.2	Simplicity	93
3.6	Commercialization of the product.....	94
3.6.1	Target Market	94
3.6.2	Marketing Mix.....	94
3.6.2.1	Product.....	94
3.6.2.2	Price	96
3.6.2.3	Place.....	96
3.6.2.4	Promotion.....	97
3.6.2.5	Ten days marketing plan to launch Eye Care.....	99
3.6.3	Maintenance	102
3.6.4	How to contact with customers	102
3.6.4.1	Business Details	102
3.6.4.2	Contact details.....	103
3.6.4.3	Online/ Social Media	103
3.6.4.4	Advisors	104
4	CONCLUSION.....	105

4.1	Image Preprocessing Findings.....	107
4.2	Research Contribution To The Field	107
4.3	Research Limitations	107
4.4	Recommendations For Future Research.....	108
REFERENCES.....		109
GLOSSARY		118
APPENDICES		122
Appendix A: Permission letter from eye hospital		122
Appendix B: Letter received from director		123
Appendix C: Permission letter from patients		124
Appendix D: Questions asked by senior consultant		125
Appendix E: Activity Diagram.....		126
Appendix F: Flow of the modules in Eye Care		127
Appendix G: Flow of the system in Eye Care		128
Appendix H: Flow of image preprocessing and segmentation module.....		129
Appendix I: High level architecture diagram		130
Appendix J: Pseudo code		131
Appendix K: Research Paper		132

List of Tables

Table 1.1: Comparison of applications	12
Table 2.1: Capture image	31
Table 2.2: Segmentation of eye	32
Table 2.3: Load face button validation	44
Table 2.4: Light sensor validation	46
Table 2.5: Calibrate button validation	47
Table 2.6: Image uploading	48
Table 2.7: Eyes detection	49
Table 2.8: Image decoding	50
Table 2.9: Eye detection in server	51
Table 2.10: Eye cropping and saving images	52
Table 2.11: Iris detection and cropping	53
Table 2.12: Image quality enhancement	54
Table 3.1: Unit testing results	65
Table 3.2: User acceptance testing results	67
Table 3.3: Business Details	102
Table 3.4: Contact details	103
Table 3.5: Online/social media	103
Table 3.6: Advisors	104

List of Figures

Figure 2.1: Waterfall Model	23
Figure 2.2: System Work Breakdown Structure	26
Figure 2.3: System Overview	28
Figure 2.4: Home page	36
Figure 2.5: Select option	37
Figure 2.6: About eye care	38
Figure 2.7: Load face	39
Figure 2.8: How to use eye care	40
Figure 2.9: Photo capturing option	41
Figure 2.10: Camera opening	42
Figure 2.11: Distance calibration	43
Figure 3.1: Image selection	59
Figure 3.2: Uploading image	60
Figure 3.3: Eye detection and eye cropping	61
Figure 3.4: Iris detection and iris cropping	62
Figure 3.5: Console output	63
Figure 3.6: Eye detection in self-camera	64
Figure 3.7: Question form	83
Figure 3.8: Questionnaire form	84
Figure 3.9: Option selection	85
Figure 3.10: Image selection option	86
Figure 3.11: How you use Eye Care	87
Figure 3.12: Loading user's image	88
Figure 3.13: Select Option interface	89
Figure 3.14: Eye Care Banner	90
Figure 5.1: Surgeries performed in 2014	105
Figure 5.2: Surgeries done from 2009 to 2014	106

List of Abbreviations

Abbreviation	Description
DLL	Dynamic Link Library
HTTP	Hyper Text Transfer Protocol
IT	Information Technology
IU	International Unit
OpenCv	Open Source Computer Vision
RGB	Red Green Blue
SD	Secure Digital Card
SDLC	System Development Life Cycle
SLIIT	Sri Lankan Institute of Information Technology
UML	Unifying Modeling Language
WHO	World Health Organization

1 INTRODUCTION

1.1 Background Context

1.1.1 Publications of facts by World Health Organization

According to the facts that have been published by World Health Organization, 246 million people are suffering from low vision [1]. About 90% of the people who have eye illnesses are living in non-developed countries with a low income [1]. Globally, uncorrected refractive errors are the main cause of moderate and severe visual impairment; cataracts remain the leading cause of blindness in middle- and low-income countries [1].

According to the above results, nowadays most of the people are suffering from eye illnesses. With the time people are looking for the easiest ways to accomplish their day to day tasks. Therefore main source of gaining information is technology. So, they are expecting more help to do their work by using modern technology.

”Eye Care” is an android application where users can check their eye illness by themselves at home. This application has mainly focused on four illnesses namely Conjunctivitis, Chalazion, Blepharitis and Cataract.

1.1.2 Eye illness explanation

Cataract is a clouding of the lens in the eye that affects vision [2]. According to the latest assessment, cataract is responsible for 51% of world blindness, which represents about 20 million people (2010) [2].

Blepharitis is a common condition that causes inflammation of the eyelids. The condition can be difficult to manage because it tends to recur [3]. In a 2009 survey, carried out in the US, 37% of patients seen by an ophthalmologist, and almost half (47%) of patients seen by an optometrist had signs of Blepharitis [4].

Conjunctivitis, also known as pinkeye, is an inflammation of the conjunctiva. The conjunctiva is the thin clear tissue that lies over the white part of the eye and lines the inside of the eyelid [5]. Approximately 1% of all primary care office visits in the United States are related to conjunctivitis [6]. Approximately 70% of all patients with acute conjunctivitis present to primary care and urgent care [7].

A Chalazion is a small bump that appears on your eyelid because of a blocked oil gland [8]. It can develop on the lower or upper eyelid, and it often disappears without treatment in about one month.

1.1.3 Image processing techniques

Image processing is a method to convert an image into digital form and perform some operations on it, in order to get an enhanced image or to extract some useful information from it [9]. It is a type of signal dispensation in which input is image, like video frame or photograph and output may be image or characteristics associated with that image [9]. Usually Image Processing system includes treating images as two dimensional signals while applying already set signal processing methods to them [9]. It is among rapidly growing technologies today, with its applications in various aspects of a business [9]. Image Processing forms core research area within engineering and computer science disciplines also. Segmentation partitions an image into distinct regions containing each pixel with similar attributes [10]. The goal of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze [11].

1.1.4 Current status of existing research

“Eye Care” application is mainly focusing on four prominent eye illnesses called Conjunctivitis, Blepharitis, Cataract and Chalazion. At present the application detects above mentioned four eye illnesses with the use of image processing and semantic web

techniques with an accuracy level of 90%. Any user can download the “Eye Care” application from Google App store after releasing the first version of the application.

1.1.5 Module description

1.1.5.1 Image preprocessing and segmentation

When a user captures an image of a face “Eye Care” application will support to adjust the distance and the contrast of the image between the user and the android phone. When a user captures the image android application will detect the pair of eyes automatically.

After capturing the image user can upload the image to the server. If user wants to browse an image he/she can select an image from the image album. After sending the image to the server, is going to detect the pair of eyes and segment in to two eyes by using a Haarcascade file. “Eye Care” application is going to detect iris and then collected images are sending to data extraction module in order to do further processing.

1.1.5.2 Data extraction and classification

Here it extracts data from the images and identify illnesses exactly using data extraction techniques. Then it will send a signal to the android application related to illness category and convert data to compare with knowledge base.

1.1.5.3 Creating rules of knowledge base

Extracted data which comes from image processing component are going to be classified into five categories namely Normal eye, Cataracts eye, Blepharitis eye, Conjunctivitis and Chalazion eye and it is going to generate rules.

1.1.5.4 Data analyzing and finalize output

“Eye Care” application prompts user to answer some questions related to the illness predicted by the results retrieved from data extraction module. In server side it creates SPARQL select queries to the knowledgebase to retrieve data. Then send the output by using a signal which contains the illness name and the stage. Then “Eye Care” application provides the final output by analyzing the output which gets from the questionnaires, output from data extraction module and knowledgebase. Final output contains the illness stage, relevant doctor’s details, near hospitals and a brief description about the eye illness with some home precautions.

1.1.6 Topic significance

Main problem at present is that Blepharitis, Chalazion, Cataract and Conjunctivitis are very common eye illnesses. There are no applications have developed to detect the above mentioned eye illnesses. Therefore “Eye Care” research group developed an android application to minimize the number of patients who are suffering from the above eye illnesses.

1.1.7 Categorization of eye applications

There are considerable amount of eye related mobile applications which has developed in the field of Information Technology. These eyes related applications can be divided in to so many categories

- Detect eye illnesses
- Visual impairment applications
- Color vision deviancy
- Vision development applications.

“Eye Care” application belongs to detect eye illnesses category. Below mentioned applications are some of the eye related applications that can find in Google app store.

1.1.8 Existing eye applications

1.1.8.1 Eye Test app

With this Eye Test app user can test their vision at home easily and totally free. After doing the tests user should be able to decide whether or not he or she should see an eye doctor (optometrist, optician) [12]. Eye Test application has only focused on testing the eye without giving any single remedy.

Main benefit of this application is that the user can test his/her eyes with seven types of eye tests which are mentioned below [12].

- Visual acuity tests
- An Ishihara Color Blindness test
- Spot the Odd! - A game that requires speed and good color vision
- 4 Amsler grid tests
- AMD test for macular degeneration
- A Glaucoma survey (How much do you know about the eye)

Main disadvantages of this application are Eye Test application has only focused on testing the eye without describing any single remedy and it has not focused on common eye illnesses [12].

With comparing to the above application, “Eye Care” application becomes unique since it detects four common eye illnesses namely Conjunctivitis, Blepharitis, Chalazion and Cataract.

1.1.8.2 Eye Care Plus app

Provided information about eye health and eye allows Eye Care Plus to come up with the best combination of eye exercises [13]. Eye Care Plus vision therapy is based on daily eye training plans and eye exercises that will help to preserve and improve user's vision.

Main advantages of this application are given below [13].

- Detects eye problems and common eye diseases.
- Gather information about eye health with free eye tests.
- In order to check knowledge user's knowledge they offer some special quizzes like Eye Quiz, Cataract Quiz, Glaucoma Awareness Quiz, and Cataract Awareness Quiz.

Disadvantage of Eye Care Plus app is that Eye Care plus application is not focusing on common eye illnesses which are prominent at present.

With comparing to "Eye Care Plus" application, "Eye Care" application becomes unique since it detects four common eye illnesses namely Conjunctivitis, Blepharitis, Chalazion and Cataract.

1.1.8.3 Vision Test 2.0 app

Users can test their eye condition and vision easily at home. User can decide whether he/she wants to see a doctor. Features of this application are listed below [14].

- Astigmatism Test
- Duo chrome Test
- Color Test
- Far field vision test
- Optician Finder

- Eye Quiz
- Eye Tips and facts

Main advantages of this application are described below [14].

- Users can test their eye condition easily at home and user can decide whether he/she wants to see a doctor.
- User receives regular reminders in order to get user's eyes tested.
- From Visual Acuity, Astigmatism and Color Tests - Vision Test will assess user's eyes in a variety of different ways.
- This application will automatically find the nearest opticians around the user.

Main disadvantage of this application is that they are not detecting eye illnesses but some eye defects like color blindness, Nearsightedness and Shortsightedness. "Eye Care" application detects four common eye illnesses namely Conjunctivitis, Blepharitis, Chalazion and Cataract when comparing to the Eye Care plus app [3].

1.1.8.4 EyeXam app

This Android app helps user to maintain healthy eyes and great vision by offering fun, easy-to-use, and patent-pending self-guided eye tests, while letting user to easily find and stay connected to eye doctor. User can also schedule an eye exam, send and receive messages to and from the doctor's office [15].

Main advantages of this application are "EyeXam" application can freely download from the Google app store, Application user can search for a local eye doctor with the GPS-based doctor finder and user can also schedule an eye exam. By using Android notification center, user can even use EyeXam as a personal mobile vision care manager [15]. Main disadvantage of EyeXam application is that it only supports Android v2.3.6 and above [4]. "EyeXam" is not focusing on common eye illnesses.

“Eye Care” application detects four prominent eye illnesses namely Conjunctivitis, Blepharitis, Chalazion and Cataract with the use of image processing and semantic web techniques when comparing to the “EyeXam” application.

1.1.8.5 Visual Acuity Test app

Free Visual Acuity test that will help user to detect problems with their vision, and identify the most common eye issues. The features of Visual acuity test application are Visual Acuity Test, Astigmatism Test, and Glasses Check [16]. Main advantages of “Visual Acuity Test” application are described below [16].

- It consists of personal vision screening to check vision improvement progress.
- General eye exams.
- Home eye tests.

Main disadvantage of EyeXam application is that it only consider about macular generation. In “Eye Care” application it detects four eye illnesses called Conjunctivitis, Blepharitis, Chalazion and Cataract when comparing to Visual Acuity Test application [16].

1.1.8.6 White Eye Detector

White Eye Detector is also an application which helps parents to detect the appearance of white eye in pictures of their children [17]. A white pupil can be a normal photographic effect in pictures (like red eye, but white). White eye can also be a symptom of multiple serious eye diseases such as retinoblastoma, pediatric cataract, and Coats' disease [17]. These diseases are often first detected by parents because the parent notices white eye in pictures of their children.

Main benefit of this application is that the application will find out pictures on your iPhone or iPod that have white eye, and will show you these pictures so that you can inspect them for yourself [17]. Main disadvantage of White Eye Detector application is that it only consider about only the appearance of the eye.

In “Eye Care” application it detects four eye illnesses called Conjunctivitis, Blepharitis, Chalazion and Cataract when comparing to White Eye Detector application [17].

1.1.8.7 Eye Exercises - Eye Care Plus

Eye Care Plus exercises address most common eye problems. There are 10 exercise categories for eye relaxation, dry eye, lazy eye and other conditions. “Eye Care Plus” application which provides 8 quizzes in order to check user’s knowledge about common eye illnesses problems and first aid becomes a main advantage.

There are 12 eye tests to track vision improvement and it offers 7 eye training plans that fits the user [18]. Main disadvantage of this application is that the user cannot detect common eye illnesses.

“Eye Care” application detects four common eye illnesses by using two main techniques called image processing and knowledge base when comparing to the above application.

1.1.8.8 Amsler grid

The Amsler Grid is a tool for identifying distortion in user’s visual field. It is a convenient eye test that anyone can use at home to monitor changes in your vision [19]. It helps to detect many problems early, while they may still be treatable, such as retinal detachments and central serous choroidopathy.

Main advantages of Amsler grid are

- An Internet connection is not required to use this app.
- User's privacy is secure because Amsler grid does not require any special permissions and it does not collect any information.
- A light weighted application
- Freely available in Google app store.

Main disadvantage of this application is that the user cannot detect prominent eye illnesses comparing to the “Eye Care” application.

1.1.8.9 Color Blind test

Color Blindness helps to identify color deficiency reveals red, green or blue color blindness or any kind of color vision weakness. This Color Blindness Test includes Ishihara Color Vision Test, Farnsworth D-15 Dichotomous Color Blindness Test and the viral Color Cube game Color Cube game [20]. By identifying the square with the different hue, and moving on to harder levels of the game, user will be able to train color vision [20].

Main advantages of the application

- Finding the exact color
- Read color coded map, chart and graph
- Color vision test

Main disadvantage of this application is that the user cannot detect common eye illnesses. When comparing to the “Eye Care” application, it detects four common eye illnesses by using two main techniques called image processing and knowledge base.

1.1.8.10 Pink Eye App

Pink eye application gives the information about the conjunctivitis symptoms [21]. Advantage of this application is that the user can get an idea about the Pinkeye illness with symptoms [21].

Pink eye app application user cannot detect conjunctivitis [21] becomes a main disadvantage of this application. When comparing to the “Eye Care” application, it detects conjunctivitis by using two main techniques called image processing and knowledge base.

There are some eye applications which give eye exercises to users of the applications. But most of the time they are giving irrelevant results [22]. EyeXam app and Visual Acuity Test app are trying to identify some eye defects but not Blepharitis, Cataract, Conjunctivitis and Chalazion. There is an application which is focusing on detecting eye illnesses called White eye detector application.

1.1.9 Researches based on image processing

A research that has done by Md. Anayet U. Patwari about Detection, Categorization and Assessment of Eye Cataracts Using Digital Image Processing has used image processing techniques in order to identify Cataract [23].

First the lens image is extracted from the patient’s photo. Then RGB image is converted to resized grey scale image for standardizing the comparison and reducing the calculation load by a third. The presence of the cataract is then detected by comparing with the mean grey intensity of healthy eyes obtained from 3 normal eyes. The image is further converted to black and white, for edge detection and circularity measurement. Image morphological feature is used for noise reduction. The filtered image is subsequently converted to binary format and its circularity compared to the circularity threshold of the mask [23].

The above mentioned research is based on Image Processing. But in Eye Care application has identified four eye illnesses by using two techniques named knowledge-based and Image Processing.

1.1.10 Comparison of existing applications

Table 1.1: Comparison of applications

Eye Applications	Eye Care Application Features				
	Detect Cataract	Detect Blepharitis	Detect Chalazion	Detect Conjunctivitis	Convey relevant illness details to user
Eye Test App	✗	✗	✗	✗	✗
Eye Care Plus App	✗	✗	✗	✗	✗
Vision Test 2.0 App	✗	✗	✗	✗	✗
Eye Exam App	✗	✗	✗	✗	✗
Visual Acuity Test App	✗	✗	✗	✗	✗
White Eye Detector	✓	✗	✗	✗	✗
Eye Care	✓	✓	✓	✓	✓

1.1.11 Preprocessing and segmentation

1.1.11.1 Image preprocessing

In imaging science, image processing is processing of images using mathematical operations by using any form of signal processing for which the input is an image, a series of images, or a video, such as a photograph or video frame; the output of image processing may be either an image or a set of characteristics or parameters related to the image [24].

There are three main types of image processing namely digital image processing, optical image processing and analog image processing while “Eye Care” application has used digital image processing.

1.1.11.2 Image segmentation

In computer vision, image segmentation is the process of partitioning a digital image into multiple segments (sets of pixels, also known as super-pixels) [25]. The objective of segmentation is to simplify and/or change the representation of an image into something that is more meaningful and easier to analyze.

In “Iris Segmentation and Detection System for Human Recognition Using Canny Detection Algorithm” they have used Iris Segmentation and Human Recognition using Canny Detection Algorithm [26]. In order to capture the image first they had captured the image and converted the image into gray level for further processing. Then they have done histogram equalization and adjust the contrast of the image. By using Hough transformation technique researchers have detected boundaries of the image and to detect edges they used Canny Edge Detecting Algorithm. Finally image is being cropped.

Pre - processing is a common name for operations with images at lowest level of abstraction both input and output are intensity images [27]. The aim of pre-processing to improve the image data or enhances some image features important for further processing. Digital images are prone to a variety of types of noise which result errors in the image acquisition process that result in pixel values that do not reflect the true intensities of the real scene.

There are so many ways that noise can be made in an image, depending on how the image is created. Examples such as if the image is scanned from a photograph made on film, the film grain is a source of noise. Noise can also be the result of damage to the film, or be introduced by the scanner itself, If the image is acquired directly in a digital format, the mechanism for gathering the data (such as a CCD detector) can introduce noise, Electronic transmission of image data can introduce noise [28].

In “Eye Detection Using Morphological and Color Image Processing” they have detected eye by using morphological and color Image Processing [29]. Next boundary region and light source reflection by eye have detected. For skin detection, they have tried to determine the exact location of the eye. Then system has trained few images for skin color manually. For this purpose they have used six-sigma technology. Here edge detection is based on eyes illumination (light dot). Image will be converted to NTSC image. For edge detection they have used morphological technique which helped to detect clear and thick boundaries. Eye dot was recognized by thresholding technique.

1.1.11.3 Histogram equalization

In order to get a good quality image “Eye Care” application has used histogram equalization. It is a method that improves the contrast in an image, in order to stretch out the intensity range [30].

1.1.11.4 Detecting face, eyes, iris from the image

In order to detect face, eyes and iris “Eye Care” application used Haar Cascade xml file. Object Detection using Haar feature-based cascade classifiers is an effective object detection method proposed by Paul Viola and Michael Jones in their paper, "Rapid Object Detection uses a Boosted Cascade of Simple Features" in 2001. It is a machine learning based approach where a cascade function is trained from a lot of positive and negative images. It has then used to detect objects in other images [31].

1.2 Research Gap

There are considerable amount of eye related application which has developed in the field of Information Technology. When it comes to visual impairment applications eyeXam [15], Amsler Grid App [19], White Eye Detector [17] can introduce as mainly running applications. But some applications are not giving accurate results [19]. An example is described below.

In vision impairments it must to maintain a distance between the patient’s eye and the Jaeger eye chart [32]. But with these eye impairments mobile applications cannot maintain the proper distance between the application and the patient’s eye. In color vision deviancy related applications, patients can identify colors. But if there is a problem in the contrast level of the phone the application will not be able catch the relevant color. In these situations application will give incorrect results.

Pink Eye app [21] will give you the idea of conjunctivitis or pink eye disease. It will make you conscious about its prevention and treatments. Here also it is just giving a brief description about the disease but not identify the disease by analyzing an image. But by this application it is to analyze the image and check whether that the user has this disease or not. Therefore these are the limitations that you can find in android applications that have developed so far.

1.2.1 Gap reduction

“Eye Care” is an android application which has developed to identify most prominent eye illnesses in modern day society in order to minimize the most common above four eye illnesses using image processing and semantic web techniques. Conjunctivitis patient get green or white discharge from eye. In Chalazion a lump will build up in upper lid or lower lid of the eye. In Blepharitis can see discharging near eye lashes. And in the Cataract can clearly identify whitish areas in lenses. To identify this eye illnesses “Eye Care” application tries to get a clear image of the infected eye by using a high pixel camera in the phone. By analyzing the image it is going to provide accurate results by identifying the illness.

1.3 Research Problem

According to the results that have been published by World Health Organization, 285 million people are estimated to be visually impaired worldwide [33]. Out of 285 million people, 39 million are blind and 246 million people have low vision [33]. Therefore according to these results about 90% people are suffering from eye illnesses in the world [33].

In a survey that has carried out in United States, 24 million Americans are affected by Cataract and they estimate 38.7 million of people will be affected by cataract by 2030 [34]. A high portion of people are seeking help from health care professionals specializing in eye care in order to get medicine for Blepharitis. In a 2009 survey, carried out in the US, 37% of patients seen by an ophthalmologist, and almost half (47%) of patients seen by an optometrist had signs of Blepharitis [35].

Conjunctivitis affects many people and imposes economic and social burdens. It is estimated that conjunctivitis affects 6 million people annually in the United States [36]. The cost of treating bacterial conjunctivitis was about \$377 million per year [36]. Many

United States health departments, irrespective of the underlying cause of conjunctivitis, require students to be treated with topical antibiotic eye drops before returning to school.

A high portion of conjunctivitis patients are initially treated by primary care physicians rather than eye care professionals. Approximately 1% of all primary care office visits in the United States are related to conjunctivitis. About 70% of all patients with acute conjunctivitis present to primary care and urgent care. It is not always possible to identify a cause for a Chalazion. However, Chalazion is more common in those with Blepharitis [37].

Above mentioned results were gained by several eye hospitals in all around the world. By using these results it can conclude that Conjunctivitis, Blepharitis, Chalazion and Cataract has become most common eye illnesses.

Most of the people who are suffering from above eye illnesses are reluctant to take medicine in hospitals due to busy schedules. Sometimes patients may be frightened of potential side effects and they may have witnessed side effects experienced by a neighbor, relation or a family member who was taking similar medication. Most of the patients cannot afford the hospital charges. Some patients may think that they do not need medicine since they are not in a severe condition. If the patients are in lack of symptoms they are trying to go for other options like getting home remedies or asking someone from social medias.

Sometimes people who are guiding in social media can be lack in knowledge which it causes some severe problems to occur. One of the main problem is the android applications which are available right now are not focusing on these four common eye illnesses. Still people have failed to develop an android application where it detects Blepharitis, Cataract, Conjunctivitis and Chalazion illnesses [15].

- **Solution Proposed**

For the above problems “Eye Care” android application is going to give a better solution where an “Eye Care” application user can check whether he/she is suffering from Conjunctivitis, Blepharitis, Chalazion or Cataract in few minutes without taking any single help from the third party. They can operate “Eye Care” application very easily since the application is more users friendly. People who are using this application they can get a clear idea whether he is in a severe condition of this illness or not. If the user is in a severe condition “Eye Care” application is going to provide details about the illness, doctors related to the illness as well as some home precautions in order to reduce the severity of the illness.

1.4 Research Objectives

1.4.1 General objective

To find out most common eye illnesses namely Chalazion, Blepharitis, Conjunctivitis and Cataract by using image processing and semantic web (Knowledge base and ontology).

1.4.2 Specific objectives

- To detect human face and eyes in android application.
- To measure the distance between user and android application and to measure the amount of light intensity of the camera in order to capture the image.
- To detect and crop eyes in server.
- To create and train the iris detection Haar-Cascade file in order to detect and crop irises.
- To enhance the quality of the image and to find a way how to send the enhanced cropped eyes and iris images to the data extraction module.

1.5 Report Overview

This document provides a detailed description of the Image Pre-Processing and Segmentation module and detailed description about how “Eye Care” application has developed by using Image Processing. People who are using the application can read this document in order to get a clear understanding about the project. Researchers who are willing to launch projects based on image processing techniques can read this document to develop further systems.

Main benefit of this document is that the reader can get a clear understanding about the image processing techniques, how image processing techniques have applied to detect illnesses and how the techniques have used to detect face features. Mainly describes all the techniques and methods used to develop “Eye Care” application. Since a complete description of the Pre-Processing and Segmentation module performed by the proposed software is provided in this document, this will help the readers to determine whether the software specified really addresses the current problem context and whether it meets the needs of the targeted community.

This document initially discusses about the similar systems and researches, their outcomes and their limitations. Then it discusses how the proposed system differs from those approaches. In this document the system and preprocessing and segmentation module is graphically represented to make the readers find it convenient to understand. It is presented in a simplified manner to address an audience with different levels of technical capabilities and relevant knowledge. In short, the purpose of this document is to provide a detailed overview of Image preprocessing and segmentation, its parameters and goals related to overall system.

1.6 Report Organization

Chapter 1: This gives a widespread introduction of research project on the whole. The introduction leads the reader from a general subject area to a particular field of research.

In literature review surveys scholarly articles, books and other information sources relevant to a particular issue, area of research, or theory, and by so doing, provides a description, summary, and critical evaluation of these works in relation to the research problem being investigated.

In research gap it has clearly mentioned the gap that need to fill with research approach. Then it has mentioned the research problem with the solution briefly. Finally it has mentioned research objectives clearly by using general objective and specific objectives.

Chapter 2: The methods section describes the rationale for the application of specific procedures or techniques used to identify, select, and analyze information applied to understanding the research problem, thereby, allowing the reader to critically evaluate a study's overall validity and reliability.

Testing has described by using a set of test cases which were designed according to a test plan. Implementation has mentioned about the methods that used to develop the application.

Finally it has mentioned about the research findings. Research findings include the principal outcomes of a research project, what the project suggested, revealed or indicated.

Chapter 3: Here results are described by using the outcomes of the “Eye Care” application. It has simply stated the findings of the research and the accuracy of the application.

Chapter 4: The conclusion is intended to help the reader understand why “Eye Care” application research should matter to them after they have finished reading the document.

2. METHODOLOGY

This section gives details on the development of methodology, system implementation and approaches that have taken on system implementation. An overview image of the system is given at the end of this document. It briefly describes the whole process of this system.

2.1 Methodology

“Eye Care” research group tried to find out a suitable system development life cycle to develop the above android application. Software Development Life Cycle is a process used by software industry to design, develop and test high quality software. The SDLC aims to produce high quality software that meets or exceeds customer expectations, reaches completion within times and cost estimates [38].

2.1.1 Referred models

Waterfall Model [39], Iterative Model [40], Spiral Model [41] and Big Bang model [42] are the models which were referred by Eye Care group in order to develop the application. The other related methodologies are Agile Model [43], RAD Model [44], Rapid Application Development [45] and Prototyping Models [39].

2.1.2 Choice of Eye Care

Out of the above system development life cycle models, “Eye Care” research group selected waterfall model to carry out the research. The waterfall method has been selected by “Eye Care” project group members after analyzing the requirements of the project and some specific reasons have been identified since there is clear compartmentalization of work and control and it is easier to set schedule for the tasks to be completed within the specific time period. The phases are not overlapping in this approach. Waterfall model is very easy to explain to the client since it is structural

approach. Here the stages and activities are well defined and it helps to plan and schedule the project very easily. Verification at each stage ensures early detection errors [46]. Each and every stage has specific deliverables. Following diagram shows the diagrammatic representation of different phases of waterfall model.

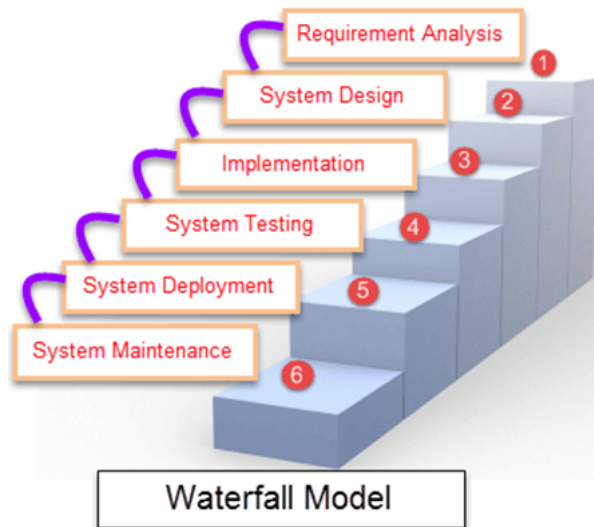


Figure 2.1: Waterfall Model

Source: The above figure is extracted from guru99 website [66].

Figure 2.1 shows one of the system development life cycle called waterfall model which was used by “Eye Care” application.

2.1.3 Background reading

Background reading was done using internet and research papers related to health sector. In this step information were gathered regarding the similar kind of systems which had been implemented earlier in the world. And also found out the main features and failures of those systems. That information has been used to compare the system that is going to be implemented with the existing systems. Finally “Eye Care” research group identified some features that should be added newly to the proposed system.

2.1.3.1 Gathering requirements

The main purpose of this step is to gather requirements and analyze them. The gathered requirements were referred by the project members and supervisors over and over again via interviews in order to be confident that the requirements are well understood, while avoiding any uncertainties. So, the following are the tasks that have done as a result of gathering requirement & analyzing phase.

- One-on-One interviews

This is the most common technique for gathering requirements. Specially “Eye Care” research group went to the National Eye Hospital and asked about the common eye illnesses from a doctor who is specialized in eye illnesses.

This discussion was well planned and “Eye Care” group members asked open-ended questions to get the interviewee to start talking and then asked questions to clarify some problems.

Questions that asked from the senior consultant doctor Prasad Colabage has mentioned in appendix D. According to his idea Blepharitis, Cataract, Chalazion and Conjunctivitis are the four main prominent eye illnesses. According to his explanation Cataract has a cloud in the eye, Conjunctivitis consists of the redness in the eye, Chalazion is a lump in the eye and Blepharitis is the redness and a green discharge in the eye. Above mentioned details are the findings which could gain from the senior consultant of National Eye Hospital. Information gained by eye hospital is shown in the appendix A, appendix B, appendix C and appendix D.

- Online-based Resources

“Eye Care” group members referred online materials such as research papers, tutorials, web articles, programming codes and e-books in order to find out valuable information. Outcomes of the online-based resources are mentioned below.

- Common eye illnesses that can be found at present are Blepharitis [47], Chalazion [48], Cataract [49] and Conjunctivitis [50].
- Symptoms of the prominent eye illnesses [51].
- Remedies for the above eye illnesses [52].
- Precautions that be taken in order to minimize the eye illness [52].
- Referring research papers

“Eye Care” members analyzed researches and articles related to image processing. A research that has done by Md. Anayet U. Patwari about Detection, Categorization and Assessment of Eye Cataracts Using Digital Image Processing has used image processing techniques in order to identify Cataract [21].

- Analyzing existing applications

Eye Test app [12], Eye Care Plus app [13], Vision Test 2.0 app [14], EyeXam app [15], VisualAcuity Test app [16], White Eye Detector [17], Eye Exercises - Eye Care Plus [18], Amsler grid [19], Color Blind test and Pink Eye App [20] are the existing applications analyzed by “Eye Care” research group to carry out the process.

2.1.4 System design

Before starting the actual coding phase, it was highly important to understand the requirements of the end user (disabled and less IT literacy people) and also have an idea of how should the end product looks like. The requirement specifications from the first phase are studied in this phase and a system design is prepared. System design helps in

specifying hardware and system requirements and also helps in defining the overall system architecture. Flow of the modules, flow of the system and flow of the image processing module is shown in appendix F, appendix G and appendix H. In this phase, “Eye Care” team members identified the interacting components that make up the system. Work break down structure of the “Eye Care” project is given below.

2.1.4.1 Work breakdown structure

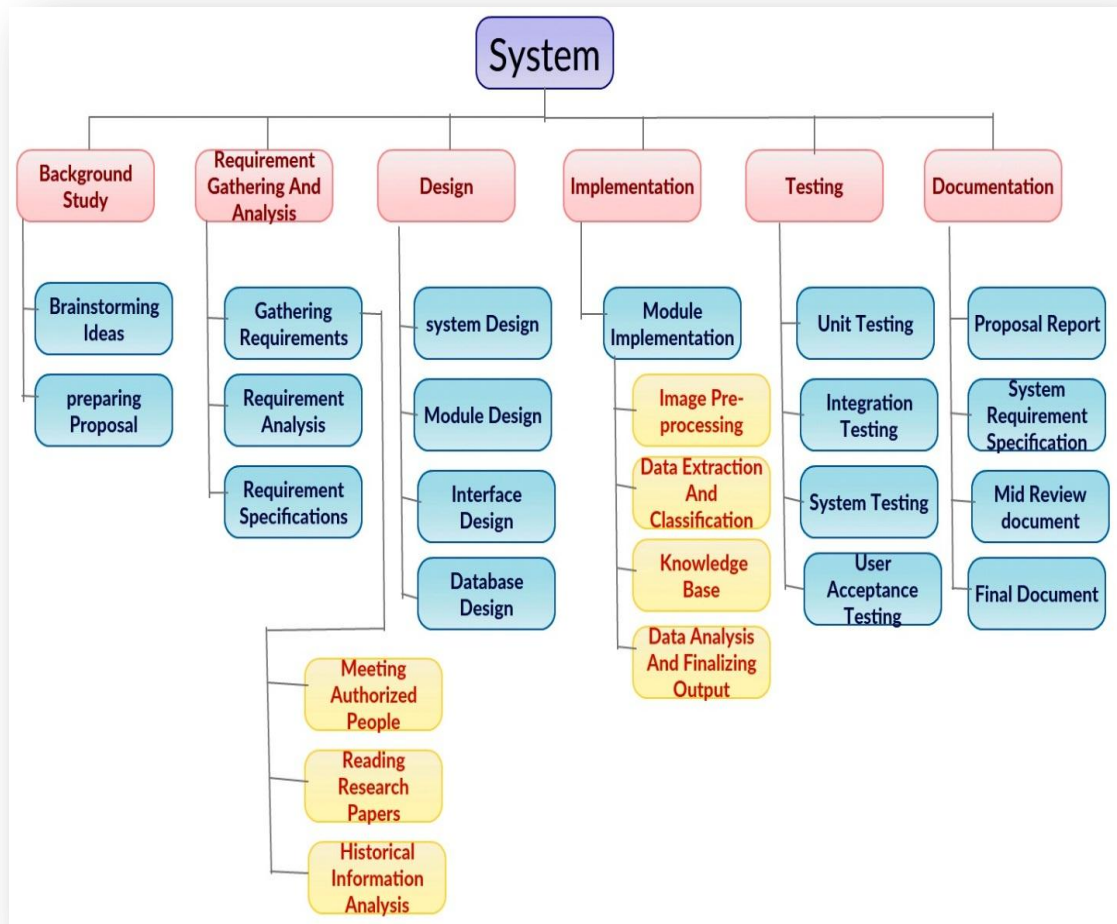


Figure 2.2: System Work Breakdown Structure

Figure 2.2 shows the system work breakdown structure of the Eye Care application.

2.1.5 Overall description

“Eye Care” android application is consist of four major modules.

- Image Pre – Processing and segmentation
- Data Extraction and Classification
- Creating rules of Knowledge base
- Data analyzing and finalizing output.

“Eye Care” is a mobile application captures user’s image and detects whether he/she is having eye illnesses. This application provides user about the health status within few seconds.

Image Pre-Processing and Segmentation module is one of the most important parts of the whole system which allows generating a quality and a cleared image with extracted eye parts such as left eye, right eye and iris. This research project is targeting eye patients.

A dedicate team of four members have continuously worked for about eleven months to achieve this task. Here user needs to provide a photo in order to carry out the process .User can select a photo from the album or he/she can capture the image by his/her own. When user selects the image from the album android application will redirect to the photo gallery where user can choose and upload a human face to the server.

If user selects capture image, it will automatically open up the camera and user can capture the image. When user captures the image the android application helps user to adjust the distance between the user and the phone and it will show the light energy of the background as well. After taking the image user can send the image to the server where it starts image pre-processing and segmentation module. In server side first it will detect two eyes and crop them as left eye and right eye. Then it will crop iris as well. Histogram equalization has used in order to enhance the quality of the image. Then the

images send to the data extraction module in order to do further processing. Activity diagram drawn for the preprocessing module is shown in appendix E.

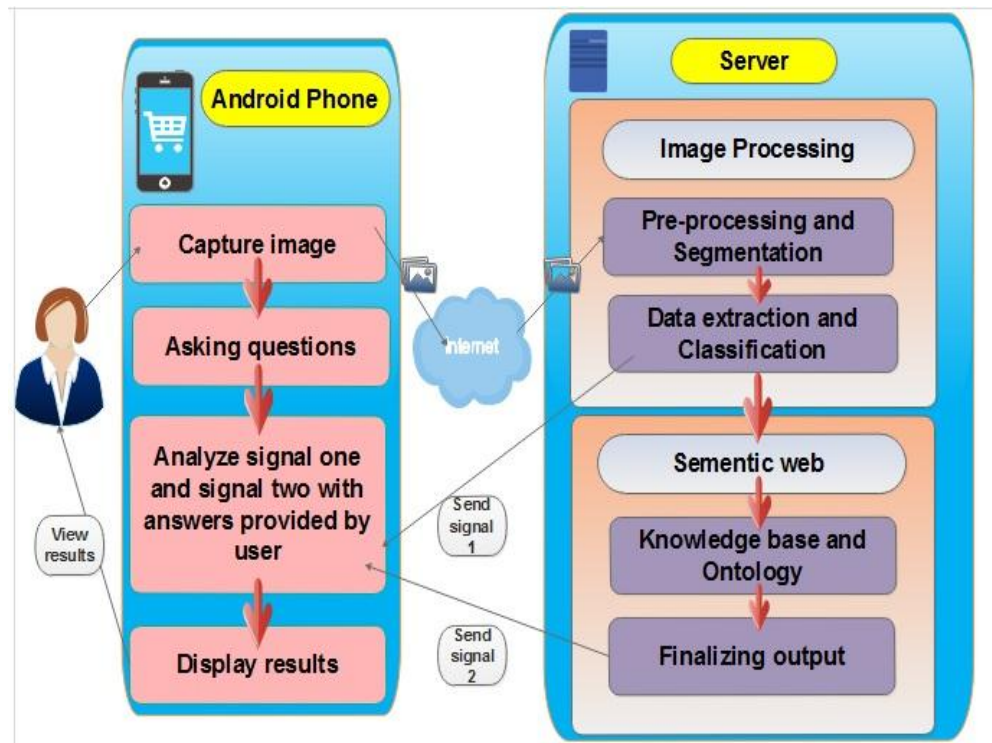


Figure 2.3: System Overview

Figure 2.3 describes the overview of the “Eye Care” application. It shows how Image preprocessing and segmentation, Data extraction, Knowledgebase and finalizing output module interacting with each other and the flow of the modules.

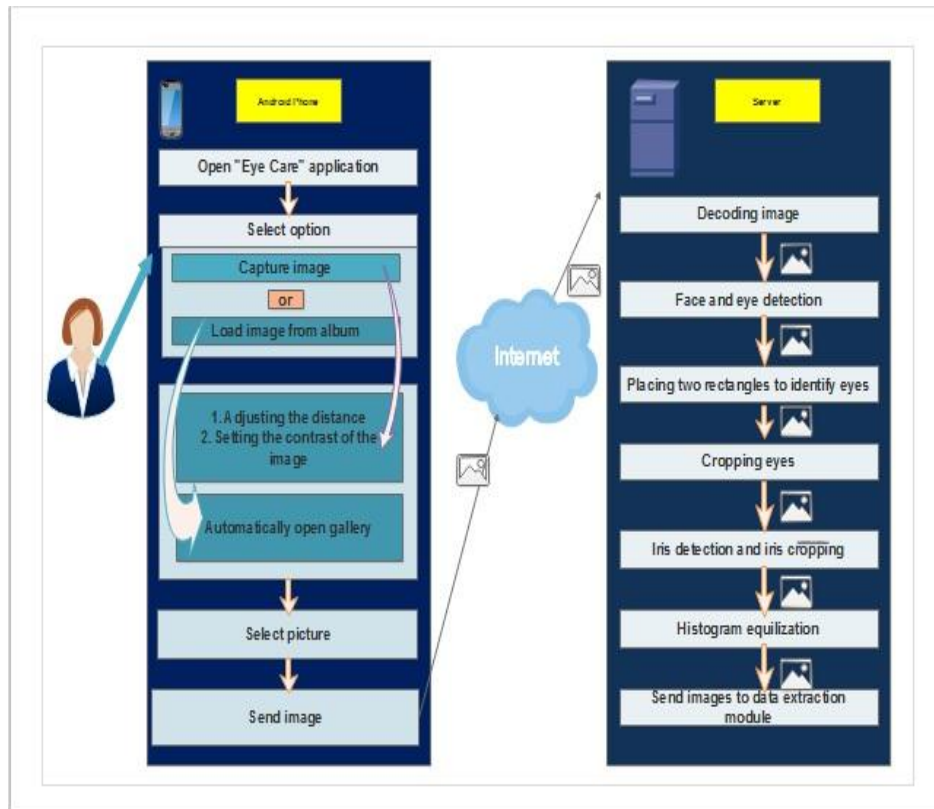


Figure 2.4: Image preprocessing and segmentation module overview

Figure 2.4 describes the overview of the image preprocessing and segmentation module. It shows how preprocessing and segmentation module interacts with the server and the android application.

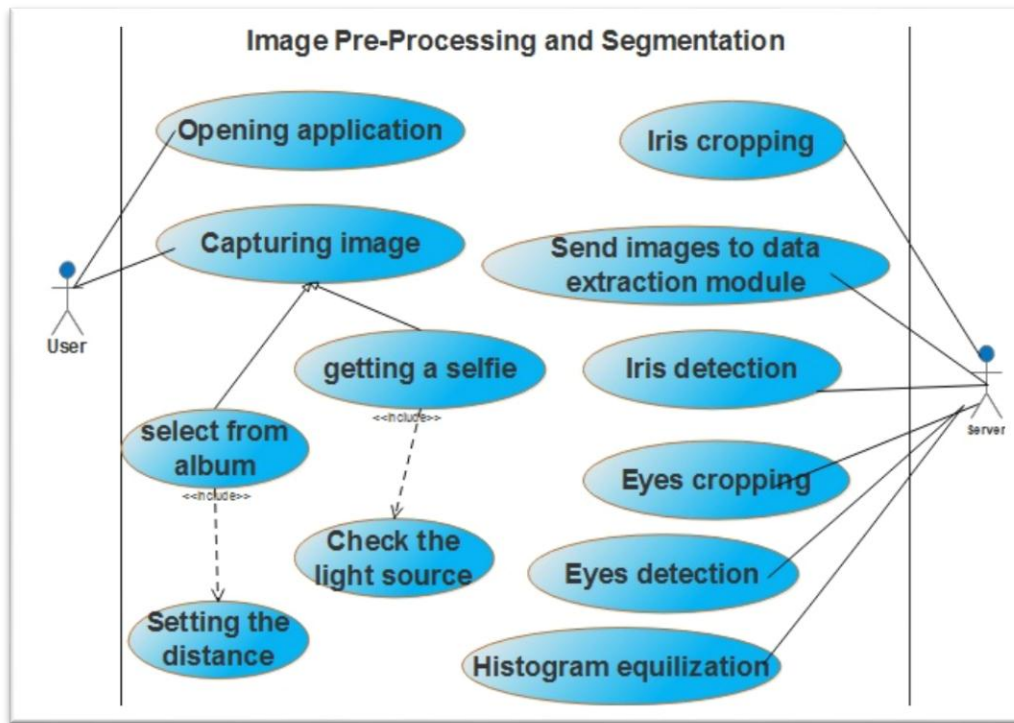


Figure 2.4: Use case diagram

Figure 2.4 use case diagram shows the interaction of image preprocessing and segmentation module with the server.

High level architecture diagram is shown in appendix I. Use case diagrams are used to gather requirements of a system [53]. Mainly it uses to get a clear understanding about the outside view of a system. It shows how external and internal factors influence the system and how it is interacting with the users.

2.1.5.1 Use case scenarios

Table 2.1: Capture image

Use Case Name	Capture Image
Use Case ID	UCS001
Description	When a user capture user's face system will ask whether the user wants to take a photo or to select the image from the photo album and captures the image according to the user preference.
Pre-Conditions	• Camera should be turn-on mode. • Before turn on the application user should provide internet service. • There must be free space in phone memory or memory card. • Before capturing an image make sure user input an image of a human face.
Post-Conditions	-
Success end condition	Successfully capture the image by detecting the pair of eyes
Failed end condition	If a face is not detected application will going to close in three minutes time.
Actors	Application User
Main Success Scenario	• System prompts the user to select auto capturing mode. • In auto-capturing mode the image will be taken by adjusting the camera distance and intensity of light. Then it will detect eyes when taking the image.

	If user wants to select the image from the photo album he/she can get an image from album.
Extensions	When capturing the image automatically the camera waits until it recognize a face or else it will close the application in three minutes.

Table 2.2: Segmentation of eye

Use Case Name	Segment the eye
Use Case ID	UCS002
Description	Here the eye is going to segment in to iris.
Pre-Conditions	- Before turn on the application user should provide internet service. - There must be free space in phone memory or memory card.
Post-Conditions	-
Success end condition	If successfully segment the image it will send to the data extraction module
Failed end condition	If the image is not segmented correctly it will show an error message.
Actors	Server
Main Success Scenario	- First it detects and crop eyes in the image. - Then it detects and crop iris. - Send segmented images to the data extraction module.
Extensions	-

When the functions of the system are taken as a whole they can be presented in the form of a use case diagram as exposed below. In addition to that, only the use cases belong to the “Image Pre-Processing and Segmentation” component that is considered as the research component in this documentation will be described in the form of use case scenarios for each relevant use case.

2.2 Implementation and Testing

2.2.1 Implementation

The implementation phase has begun as soon as the system development comes to the end of the cycle. In this phase the system was developed according to the design document. Each component of the design is implemented as program module. Basically the system was developed using java language. End of this phase is a fully functional system which is delivered successfully.

Image preprocessing and segmentation module has been divided in to two main sections as Server application and android application. Android application is used to capture the user’s image which has been developed by using java language where any user can capture an image and uploading it to the server to do further processing. Preprocessing and segmentation is done in java server.

Java web server is developed using java language and the server is made up of “Apache Tomcat” [54]. A major task of the server is to do the image processing and data extraction modules. OpenCv [55] open source library was used to implement image processing. Web server has the capabilities of receiving images from client, saving images in server, segmenting image, face detection, eye and iris detection and histogram equalization.

There are two options to capture an image by the user.

1. Image capturing and server connection

When user captures an image he/she can adjust the distance between the user and the android phone. There is a light sensor to enhance the quality of the image. If the above two properties maintain correctly user can capture the image.

In order to handle light energy first user must check whether the light sensor is available in the phone. There are inbuilt functions available to access smart phone physical resources. Using inbuilt functions application can retrieve existing energy of light which is indicating by Luminas (lu) where it measures the light spectrum. 1000lu was the minimum light source in order to take a quality image. If the light source is lower than 1000lu the application is not allowing to capture the image. In order to carry out the above process the application is using a thread.

When user captures the image, application will detect two eyes and it is viewing the eyes in rectangle. Haar cascade files have used to detect eyes. When user open the camera system immediately and reclusively checks the distance between the user and the android phone. According to the calculations the best distance range was 20U to 25U where U depicts the units. When user comes up with the correct distance “open camera” button will be visible in order to capture the image.

2. Choosing an image from gallery

“android-async-http-1.4.4” [56] library was used to send the image to the server. This library is capable of sending and retrieving data using HTTP. It also can encode and decode an image. Using this library the image is converted to a string and then sends to the server where the server is going to decode the image by using the retrieved string form android application.

In order to develop the application “Eye Care” has used java web server where it needed to host by using a static internet protocol address. It will cost more amount of money to get a static internet protocol address. Router was used as a solution to the above problem. Then the android application and server machine both connected to the router where it assigns internet protocol address manually. When client asks to connect with the server the router will by pass to web server as all the devices are in a same network. When a client connects to the web server it will grab all the information about the client such as internet protocol address, port number, image name, image size and image. Then it fires the .jsp file and starts to run all the functions in the server. “CV_HAAR_DO_CANNY_PRUNING” [57] algorithm was used to detect eyes and calculate number of eyes.

“CascadeClassfire” algorithm [58] was used to crop eyes. Then the cropped eyes were scaled to 150*150 resolution image and saved them as left eye and right eye. In order to detect irises, a newly created Haar cascade file was used. Histogram equalization was used to enhance the quality of the image. If the image consists with low contrast information, the system is capable of reconstructing the image.

Finally the collected data is sent to the “Data Extraction module” in order to do the further processing. User interface of the “Eye Care” application are shown below.

2.2.2 User interfaces

2.2.2.1 Home screen



Figure 2.4: Home page

Figure 2.3 depicts the main page of the Eye Care application. User must click Start to check your eye button in order to start the application or else the user can select exit button in order to quit from the application.

2.2.2.2 Option selection



Figure 2.5:Select option

Figure 2.2 shows three options. If user wants to check eyes he/she can click Test your eye button. User can select How to use eye care button in order to get the information about how user can use the application. About eye care gives a brief description about the application.

2.2.2.3 About Eye Care

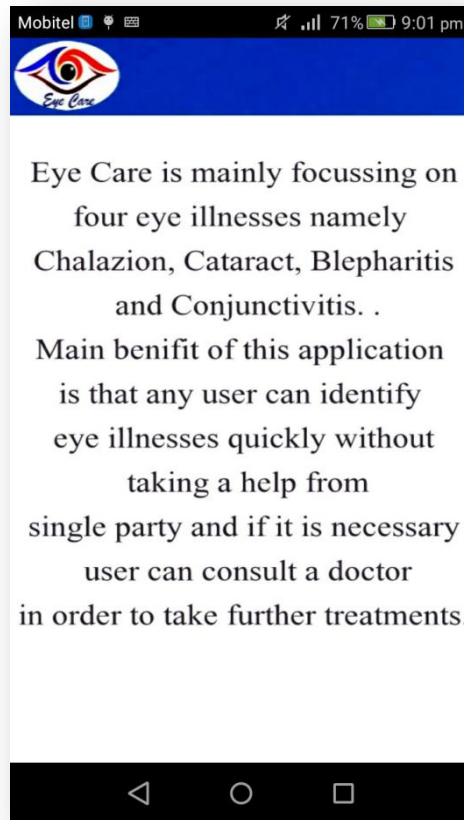


Figure 2.6: About eye care

Figure 2.5 shows a brief description about the eye care application.

2.2.2.4 Uploading face

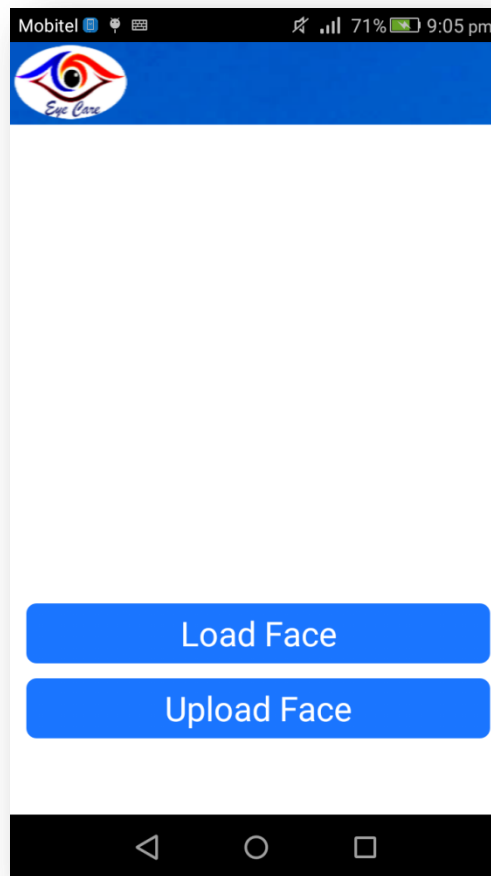


Figure 2.7: Load face

Figure 2.6 the image selection from the album page. If user selects select photo from album it ask the user to load face form album. User can upload the face to their server by clicking Upload face button.

2.2.2.5 How to use Eye Care

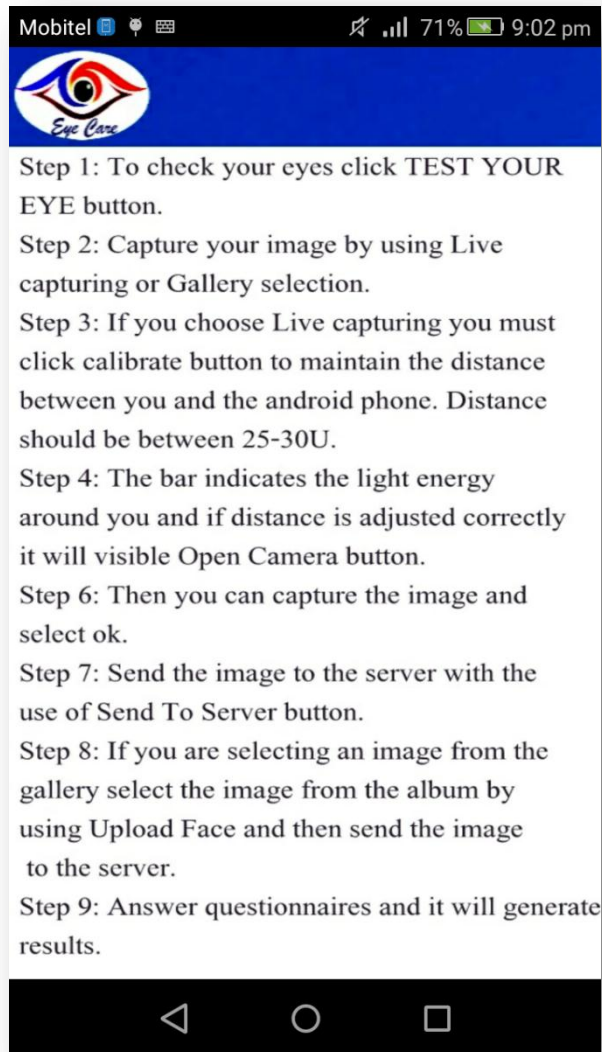


Figure 2.8: How to use eye care

Figure 2.7 shows how to use Eye care application.

2.2.2.6 Photo capturing

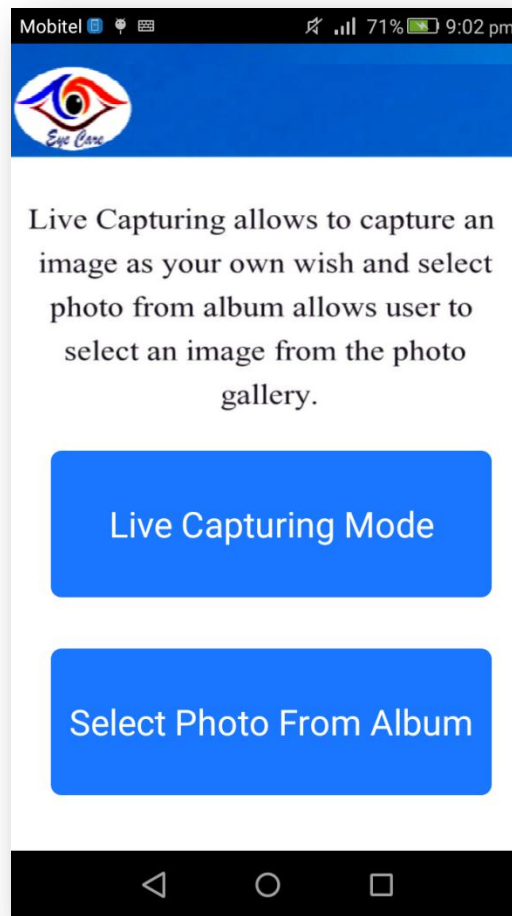


Figure 2.9: Photo capturing option

Figure 2.8 shows two options where user can select any option according to their preference. If user selects Live capturing, it captures the image by opening the camera while the other option allows user to select a photo from album and then upload the image to the server.

2.2.2.7 Camera opening

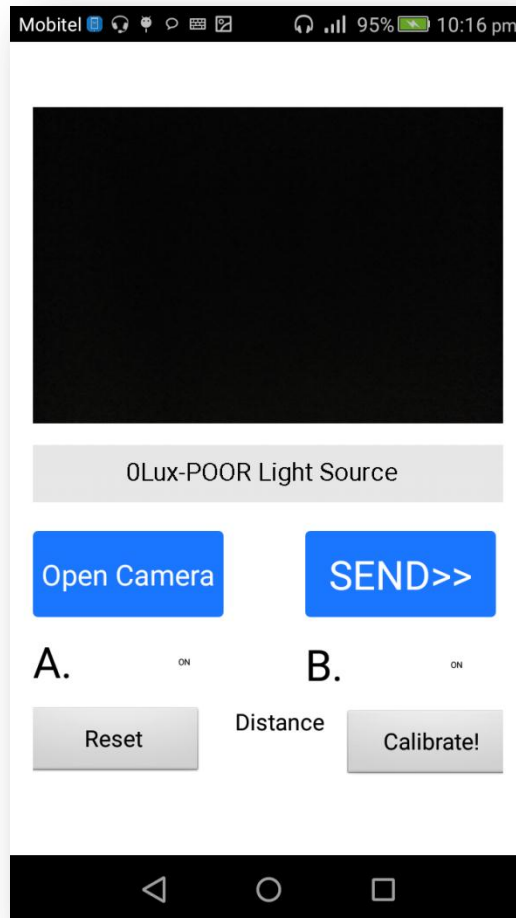


Figure 2.10: Camera opening

Figure 2.9 shows the camera which opens when user click live capturing option.

2.2.2.8 Distance calibration



Figure 2.11: Distance calibration

Figure 2.10 depicts live capturing interface. Here the user can adjust the distance between user and the android phone until it gives a beep sound. It shows the light intensity around the environment and by using A and B buttons it detects eyes. In order to calculate the distance user needs to click calibrate button and it shows the distance in U (Units).

2.2.3 Testing

2.2.3.1 Unit testing

The designing process is divided into smaller parts known as units, and unit testing needs to be carried out for each of these divisions individually. It means on receiving system design documents, the work is divided in modules/units and actual coding is started. The system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality. This is referred to as unit testing.

2.2.3.2 System testing

Once the units which system is first divided into units which are developed and tested for their functions are declared to be flaw free, they are integrated into the final system and then this system is tested to ensure proper integration and compatibility between the various units. Waterfall model in testing can only be done by dividing up the coded program into various manageable parts. As specified above, these units are integrated into a complete system during integration phase and tested to check if all modules/units coordinate with each other and the system as a whole behaves as per the specifications.

2.2.3.3 Test cases

Table 2.3: Load face button validation

Test case ID	TC001
Test priority	Medium
Module name	Image Pre-Processing and Segmentation
Test title	Validating load face button.
Description	When user clicks “Load face” button it must load photo album from the phone and then it must load the selected image to the image

	view.
Pre-conditions	User must select a proper image. User must select a human face. User must select an individual person's photo.
Test steps	1. Select and click on "Eye Care" icon. 2. Click "Start to check your eye" button. 3. Click "Test your eye" button. 4. Click "Select photo from album" button. 5. Click "Load face" button. 6. Select a proper image. 7. Then click ok button.
Test data	-
Expected results	1. Opening image gallery. 2. Loading image to the image view.
Actual results	3 Successfully opening image gallery. 4 Successfully loading image to the image view.
Status (Pass/Fail)	Pass
Post-conditions	Click "upload face" to send the image to the server.

Table 2.4: Light sensor validation

Test case ID	TC003
Test priority	Medium
Module name	Image Pre-Processing and Segmentation
Test title	Validating light sensor
Description	Phone must auto detect the light intensity around the environment
Pre-conditions	Light sensitive value should be above zero value
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Click “Live capturing” button. 5. Bringing the phone near a light source and check whether the light sensor is working properly.
Test data	-
Expected results	Showing different values in the light meter when a phone is nearing to a light source
Actual results	Successfully changed values in the light meter.
Status (Pass/Fail)	Pass
Post-conditions	-

Table 2.5: Calibrate button validation

Test case ID	TC004
Test priority	Medium
Module name	Image Pre-Processing and Segmentation
Test title	Validating calibrate button
Description	When user clicks calibrate button it must measure the distance between the user and the phone
Pre-conditions	User must capture a human image.
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Click “Live capturing” button. 5. Click calibrate button to measure the distance 6. Adjusting the distance between user and phone until open camera button appears in the self_cam page.
Test data	Live human face
Expected results	Calculating the distance between user and the phone
Actual results	Successfully calculated the distance between user and the phone
Status (Pass/Fail)	Pass
Post-conditions	-

Table 2.6: Image uploading

Test case ID	TC005
Test priority	Medium
Module name	Image Pre-Processing and Segmentation
Test title	Uploading image to the server
Description	After capturing the photo it must send to the server
Pre-conditions	User must capture an image.
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Click “Live capturing” button. 5. Click calibrate button. 6. Adjusting the distance between user and phone until open camera button appears in the self_cam page. 7. Then click open camera button. 8. Capture image 9. Click ok 10. Then click send button in order to send the image to the server
Test data	728*1024 resolution image.
Expected results	Uploading image to server
Actual results	Successfully uploaded the image to the server.
Status (Pass/Fail)	Pass
Post-conditions	-

Table 2.7: Eyes detection

Test case ID	TC006
Test priority	Medium
Module name	Image Pre-Processing and Segmentation
Test title	Detecting eyes successfully
Description	When user capturing a photo it will auto detect eyes
Pre-conditions	User must capture a human image.
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Click “Live capturing” button. 5. Adjusting the distance between user and phone until open camera button appears in the self_cam page.
Test data	Live photo
Expected results	Detect two eyes
Actual results	Successfully detected two eyes
Status (Pass/Fail)	Pass
Post-conditions	-

Table 2.8: Image decoding

Test case ID	TC006
Test priority	High
Module name	Image Pre-Processing and Segmentation
Test title	Checking whether the image is decoding properly.
Description	Checking whether the image is decoding properly in the server side which sent from the android application.
Pre-conditions	• Android application must send an image and cannot send a low resolution image to the server. • Before sending the image to the server make sure that server is up.
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Select user’s preference 5. Capture image. 6. Send the image to the server. 7. Go to the folder where images are saving when it sending images from android application to the server. 8. Compare the image that has send to the server with the image received to the server.
Test data	Human image
Expected results	Decoding image properly.
Actual results	Decoding image successfully.
Status (Pass/Fail)	Pass

Post-conditions	-
-----------------	---

Table 2.9: Eye detection in server

Test case ID	TC007
Test priority	High
Module name	Image Pre-Processing and Segmentation
Test title	Detecting eyes in the server and saving them properly in the image folder.
Description	Detecting eyes of the image which was sent by android application
Pre-conditions	Android application must send a human face and before sending the image to the server make sure that the server is up
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Select user’s preference 5. Capture image. 6. Send the image to the server. 7. Go to the folder where images are saving when it sending images from android application to the server.
Test data	Human image
Expected results	Eyes are marking as two rectangles and saving them in the folder.
Actual results	Successfully eyes are marking as two rectangles and saving them in the folder.
Status (Pass/Fail)	Pass
Post-conditions	-

Table 2.10: Eye cropping and saving images

Test case ID	TC008
Test priority	High
Module name	Image Pre-Processing and Segmentation
Test title	Cropping eyes and saving them properly in the image folder.
Description	Cropping eyes of the image which was sent by android application
Pre-conditions	• Android application must send a human face. • Before sending the image to the server make sure that the server is up
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Select user’s preference 5. Capture image. 6. Send the image to the server. 7. Go to the folder where images are saving when it sending images from android application to the server.
Test data	Human image
Expected results	Eyes are cropped and saving inside the image folder
Actual results	Successfully eyes are cropped and saving inside the image folder.
Status (Pass/Fail)	Pass
Post-conditions	-

Table 2.11: Iris detection and cropping

Test case ID	TC009
Test priority	High
Module name	Image Pre-Processing and Segmentation
Test title	Detecting iris, Cropping iris and saving them properly in the image folder.
Description	Detecting iris using Haar Cascade file, Cropping iris of the image which was sent by android application
Pre-conditions	• Android application must send a human face. • Before sending the image to the server makes sure that the server is up.
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Select user’s preference 5. Capture image. 6. Send the image to the server. 7. Go to the folder where images are saving when it sending images from android application to the server.
Test data	Human image
Expected results	Iris are detected, cropped and saving inside the image folder
Actual results	Successfully iris are detected, cropped and saved inside the image folder.
Status (Pass/Fail)	Pass

Post-conditions	-
-----------------	---

Table 2.12: Image quality enhancement

Test case ID	TC010
Test priority	High
Module name	Image Pre-Processing and Segmentation
Test title	Enhancing the image quality.
Description	Image quality is going to be enhanced for further processing.
Pre-conditions	• Android application must send a human face. • Before sending the image to the server make sure that the server is up
Test steps	1. Select and click on “Eye Care” icon. 2. Click “Start to check your eye” button. 3. Click “Test your eye” button. 4. Select user’s preference 5. Capture image. 6. Send the image to the server. 7. Go to the folder where images are saving when it sending images from android application to the server.
Test data	Human image
Expected results	Image contrast must be greater than the actual image which was sent by the android application.
Actual results	Image contrast is greater than the actual image which was sent by the android application.
Status (Pass/Fail)	Pass
Post-conditions	-

2.3 Research Findings

Eye Care application is a combination of mobile application and server. This research has used two main techniques named Image Processing and Knowledge Base in order to identify four eye illnesses called Cataract, Blepharitis, Conjunctivitis and Chalazion. In order to achieve the goal first the concept image processing was used. But while working with image processing several problems were raised.

2.3.1 Image processing technique issues

- How to detect eyes using Haar Cascade file
- How to crop eyes by using Haar Cascade file
- How to write a Haar Cascade file for iris detection and iris cropping.

There is less number of researches based on image processing to identify eyes and to crop eyes. And still there is no Haar Cascade file is written in order to detect iris and to crop iris. Preprocessing and segmentation module consists of two major parts namely android application and server.

2.3.2 Image processing module findings

2.3.2.1 Configuration of android application

When android application developed it has occurred lots of compilations errors, runtime errors, runtime exceptions and dynamic linking errors. Major problem was raised in simulators which were installed in android studio. When simulator runs, periodically it tries to crash the running program as well as the speed of the simulator is too slow. In order to overcome the above problems a suitable driver was installed in the pc, so that it was easy to run or debug the android application without wasting time and effort.

2.3.2.2 Accessing smart phone memory

There should be a proper way to retrieve images from smart phone internal memory or else from the SD card. In order to achieve the above task first the android application must get the permission from android manifest to access the internal memory or SD card. After retrieving the image data from the internal memory/SD card, the data must be stored in a temporary location so that the image can be loaded in to the camera surface view easily.

2.3.2.3 Exchanging images

Communication libraries were used in order to exchange information between android application and server. There were lots of libraries to exchange information. In order to get the most suitable libraries to exchange data and images at first some of the libraries were downloaded and checked the accuracy and time performance. “android-async-http-1.4.4” [56] was the best library as it facilitate image encoding and decoding. Servlets and jsp were also used to complete the above task.

2.3.2.4 Distance calibration

Distance between user and the android application must be calculated in order to maintain the quality of the image. Here the android phone will guide the user to set the distance correctly. When user zoom the camera, it changes the distance between two eyes. When the distance is satisfied user can capture the image.

2.3.2.5 Measuring the light source

This option helps the client to get a quality image with recommended light energy. Here it uses smart phone light sensor to measure the light energy. If the light energy is lower than the recommended range (1000lu) android phone is not allowing to capture the image.

2.3.2.6 Detection of iris

There are numerous Haar cascade files to detect eyes, faces and various objects. But still researchers have failed to develop a Haar cascade file to detect iris in java language. In “Eye Care” application it has developed a Haar cascade file for iris detection.

Pseudo code of the android application is shown in appendix J.

3 RESULTS AND DISCUSSION

3.1 Results

This chapter covers the results achieved from the research project and what were the new approaches found to address further research in the undergraduate context. An acceptance test can be understood as a way to check if a previously defined “contract” between the developer and the customer is still on track. Running those acceptance tests also ensures that no requirement change has happened in the meantime and that everything is as it should be to satisfy the customer.

3.1.1 Acceptance testing importance

- Capturing user requirements in a directly verifiable way.
- Identify problems which are missed by unit testing or integration testing.
- Providing an overview on how the system is working.

3.1.2 Image preprocessing module outcomes

3.1.2.1 Image selection from album

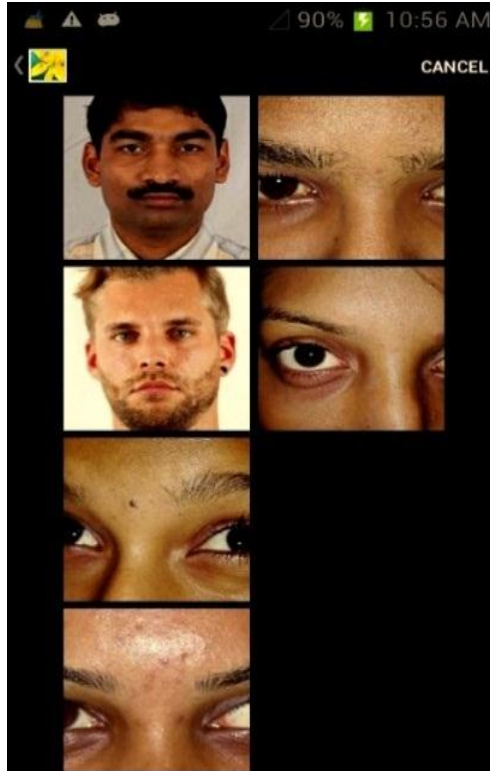


Figure 3.1: Image selection

Figure 3.1 shows how user selects a photo from album in the android phone.

3.1.2.2 Uploading image

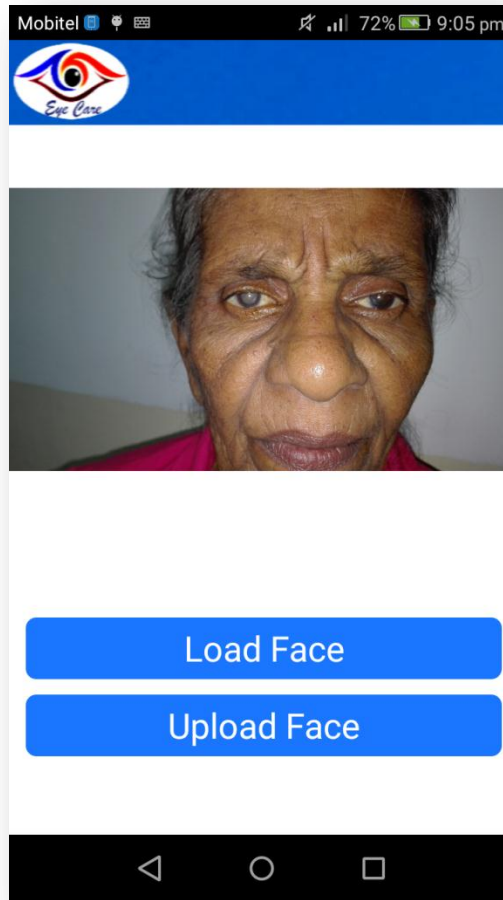


Figure 3.2: Uploading image

In figure 3.2 it shows how user uploads the image to the server. When user loads the image from the gallery by clicking Load Face button then he/she must click the upload face in order to send the image to the server.

3.1.2.3 Eyes detection and cropping

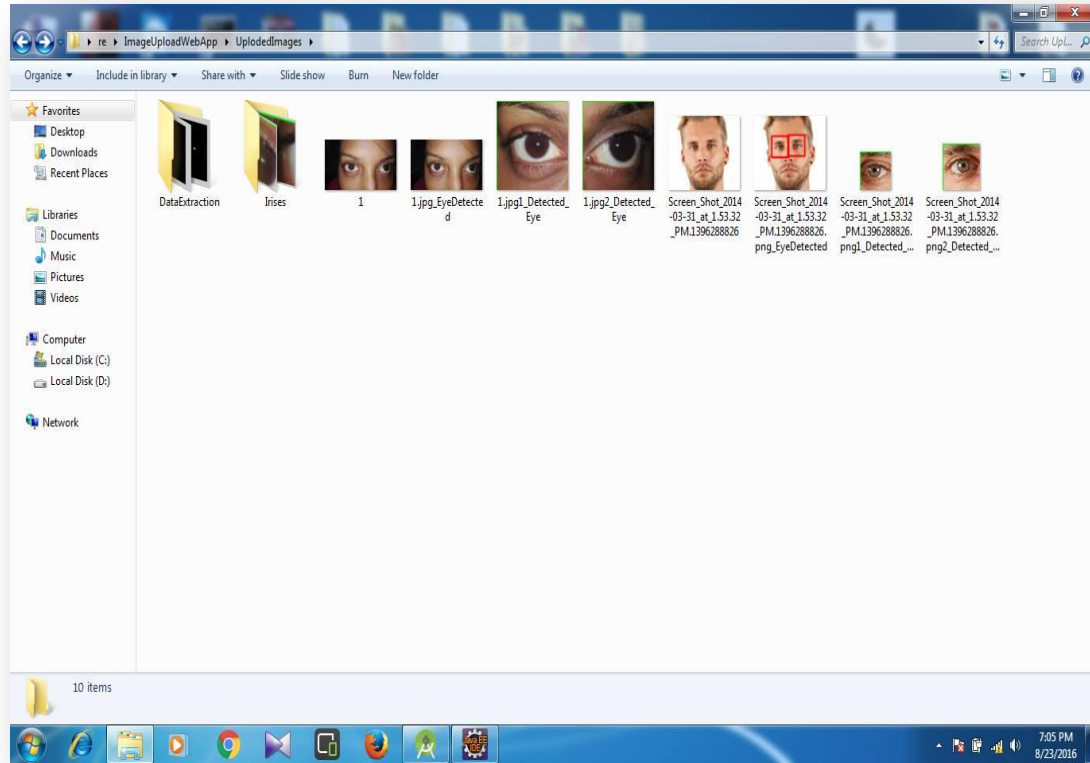


Figure 3.3: Eye detection and eye cropping

Figure 3.3 shows the images of eyes and iris which were detected in server application. Here the eyes are detected, cropped and have saved in folders.

3.1.2.4 Iris detection and cropping

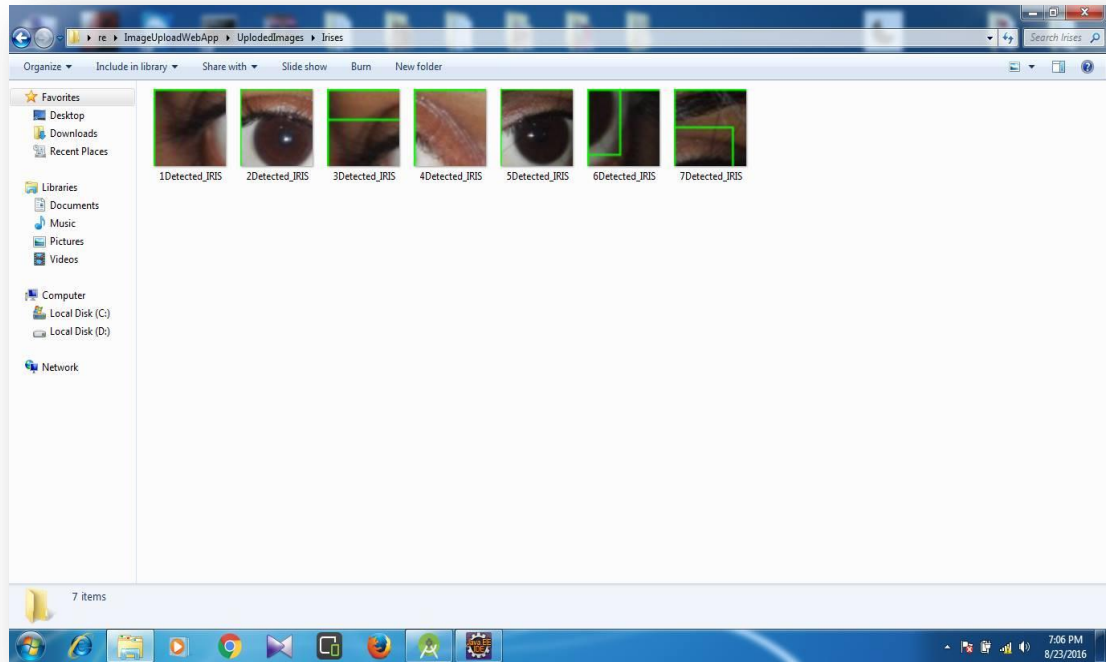


Figure 3.4: Iris detection and iris cropping

Figure 3.4 depicts the images of cropped irises.

3.1.2.5 Results output

```

Tomcat v8.0 Server at localhost [Apache Tomcat] C:\Program Files\Java\jdk1.8.0_20\bin\java.exe (Aug 23, 2016, 7:01:48 PM)

----- CLIENT CONNECTED !! -----
----- 192.168.201.45 -----
----- 39039-----

Filename: Screen_Shot_2014-03-31_at_1.53.32_PM.1396288826.png
Image Tested !!
Image upload complete, Image Saved Sucessfully !!
Image Preprocessing Module Started !!
Total Deteted Eyes : 2

Running Eye Cropping !!
Detected 2 Eyes
Eyes Cropped!!
Cropped Eye name:C:\Users\Mango\Desktop\re\ImageUploadWebApp\UplovedImages\Screen_Shot_2014-03-31_at_1.53.
Cropped Eye name:C:\Users\Mango\Desktop\re\ImageUploadWebApp\UplovedImages\Screen_Shot_2014-03-31_at_1.53.

IRIS Cropping Started !!

Running IRIS Cropping For EYE 01 !!
Detected 0 Samples
  
```

Figure 3.5: Console output

When a client connected to the server it displays in the console by mentioning about the details of the client, images, eye detection, eye cropping, iris detection and iris cropping.

3.1.2.6 Eyes detection in self-camera



Figure 3.6: Eye detection in self-camera

Figure 3.6 depicts live capturing interface. Here the user can adjust the distance between user and the android phone until it gives a beep sound by clicking calibrate button. It shows the light intensity around the environment and by using A and B buttons it detects eyes. In order to calculate the distance user needs to click calibrate button and it shows the distance in U (Units).

3.2 Discussion

According to the test results, it clearly shows that the system can

- Detect eyes when capturing an image by the user
- Adjust the distance between the user and the android phone
- Display the light intensity
- Detecting and cropping eyes in server application
- Detecting and cropping iris in server application
- Enhancing the image quality by using histogram equalization

As it is mentioned earlier, image processing techniques were used in order to implement the above functions. The following table shows the results gained by using unit testing. Unit testing is a level of software testing where individual units/ components of software are tested. The purpose is to validate that each unit of the software performs as designed [59].

Table 3.1: Unit testing results

Test number	Function name	Number of inputs	Number of correct outputs	Accuracy (Number of correct outputs/ Number of correct inputs)*100%
1	Open up the image gallery when user clicks “Load Face”	20	20	100%
2	Uploading image to the server	50	50	100%
3	Opening the camera when the	50	50	100%

	application detects two eyes and distance set correctly			
4	Detecting eyes of the user by using green color rectangles	50	50	100%
5	Calculating the distance between user and the android phone	50	50	100%
6	Generating the light intensity value correctly	10	10	100%
7	Detecting eyes in server application	150	140	93.33%
8	Detecting iris in server application	150	130	86.66%
9	Cropping eyes in server application	150	145	96.67%
10	Cropping iris in server application	150	120	80%
11	Histogram equalization	150	150	100%

The following table shows the results gained by using user acceptance testing. Testing an application prior to customer delivery for functionality and usability using real-world scenarios which resemble how the application will be employed by the end user [60].

Table 3.2: User acceptance testing results

Test number	Function name	Number of inputs	Number of correct outputs	Accuracy (Number of correct outputs/ Number of correct inputs)*100%
1	Open up the image gallery when user clicks “Load Face”	20	20	100%
2	Uploading image to the server	50	50	100%
3	Opening the camera when the application detects two eyes and distance set correctly	50	50	100%
4	Detecting eyes of the user by using green color rectangles	50	48	96%
5	Calculating the distance between user and the android phone	50	50	100%
6	Generating the light intensity value correctly	10	9	90%
7	Detecting eyes in server application	150	120	80%
8	Detecting iris in server application	150	100	66.67%
9	Cropping eyes in server application	150	145	96.67%
10	Cropping iris in server application	150	100	66.67%
11	Histogram equalization	150	150	100%

Following is the accuracy rate of the preprocessing and segmentation module which was gained by testing 50 images. This is considered as system testing. System Testing is a level of the software testing where complete and integrated software is tested [61].

No of input images: 150

No of correct results: 125

Accuracy = $(125/150)*100\%$

= 83.33%

According to the above answer it can be concluded that the accuracy of the whole image processing module is about 83.33%. But when comes to user acceptance testing, iris detection and iris cropping have the lowest results as 66.67%.

3.2.1 Limitations

3.2.1.1 Limitations of iris detection and iris cropping

In iris detection and iris cropping it detects and crop irises accurately. But when images are saving in the folder it is going to save with some garbage images. Here the system cannot identify the iris directly. But when user checks the image folder he/she can view the irises photos clearly.

If a user use a camera with a poor resolution it is hard to detect eyes and iris in the server application. If the image contains a group photo server is falling down. Therefore user needs to enter an individual face image.

3.2.1.2 Quality of the image

The quality of the image in Eye Care application is maintained with the use of two main functions in the Eye Care application. They are distance calibration and light energy adjustment. In order to enhance the quality of the image Eye Care application supports the user to capture the image by distance calibration method. Here the user can adjust the distance between user and the android phone to capture a better image. If the distance is between 25U to 30U user can capture the image. If the quality of the camera is low then the accuracy of the results will become low.

3.2.1.3 Energy of light

User needs to capture the image where the light energy is high. Eye Care application will indicate the light energy around the user. So that the user needs to capture the image in a place where it contain light more than 1000 units which displays in the android application. If user captures the image in a darker place Eye Care application tends to provide the results with a low accuracy.

3.2.2 Reasons to lower the accuracy of iris detection and cropping

In order to train a Haar cascade file it needs about 60 hours to create the Haar cascade. It is a huge time consuming work since the Haar cascade file must train with the use of about 2000 images of positive images and 5000 images of negative images. The main disadvantage of this tool is if the program stuck in the middle it should start from the beginning stage.

3.3 Standards

3.3.1 Coding standards

It is important for the Eye Care development team to select proven design techniques and good coding standards in order to build successful software. The selection of coding standards results in code consistency, which makes it easier to understand, develop and specially to maintain the application. In addition by being aware of and following the right coding techniques at a granular level, the programmer can make the code more efficient and performance effective.

3.3.1.1 Beginning Comments

All source files are beginning with the header as follows. It is carrying the Title, Date in mm/dd/yy format and the other information. Example is shown below.

Example:

```
/*
```

Title: Image processing and segmentation module

Date: 03/28/2016

```
page language="java" contentType="text/html; charset=ISO-8859-1"
```

```
pageEncoding="ISO-8859-1"
```

```
*/
```

3.3.1.2 Indentation

In computer programming languages, indent is mostly using to format program source code in order to improve the readability of the user. When Eye Care develops the application it used four spaces as the unit of indentation. An example is shown below.

```
public void EyeDetection ( String imgName ) {

    String
filePath="C:\\Users\\Mango\\Desktop\\re\\ImageUploadWebApp\\UplodedImages\\";

    String XML_FILE =
"C:\\Users\\Mango\\Desktop\\re\\ImageUploadWebApp\\CascadeFiles\\haarcascade_eye
.xml";

    IplImage img = cvLoadImage(filePath+imgName);

    detect(img,imgName);

}
```

3.3.1.2 Left and Right braces

Eye care uses starting brace at the end of the conditional and the ending brace on a separate line and aligned with the conditional. Example is shown below.

```
if(imgEncodedStr != null){

    ManipulateImage.convertStringtoImage(imgEncodedStr, fileName);

    System.out.println("Image upload complete, Image Saved Sucessfully !!");

}
```

```

        PreProcessing obj=new PreProcessing();

        obj.ImgPro(fileName);

    } else{

        System.out.println("Inside else");

        System.out.print("Image is empty");

    }

```

3.3.1.3 Wrapping lines

Eye Care does not use more than 80 characters in a single line. It is broken by an operator. Example is shown below.

```

FileOutputStream imageOutFile = new
FileOutputStream("C:\\Users\\Mango\\Desktop\\re\\"
                + "ImageUploadWebApp\\UploadedImages\\" + fileName);

```

3.3.1.4 White Spaces

Blank lines improves the readability by setting of sections that are logically related. Eye Care developers used two blank lines between class and interface definitions and one blank line used between starting of methods, before a block or a single line comment, between logical section inside a method and before and after comments. Example is shown below.

```

public void EyeDetection(String imgName){

    String
filePath="C:\\Users\\Mango\\Desktop\\re\\ImageUploadWebApp\\UploadedImages\\";

    //body

}

public void detect(IplImage src,String imNm){

    //body

}

System.out.println("Total Deteted Eyes : "+total_Eyes);

cvSaveImage(path+imNm+"_EyeDetected.jpg", src);

}

```

3.3.1.6 Blank Spaces

- Each keyword followed by a parenthesis is separated by a space. Example is shown below.

```
detect ( img , imgName );
```

- All binary operators except a period ‘.’ Is separated from their operands by spaces. Example is shown below.

```
Imgproc.rectangle ( image, new Point ( rect.x, rect.y ), new Point ( rect.x +
rect.width, rect.y + rect.height ) , new Scalar ( 0, 255, 0 ) );
```


- The expressions in a for statement are separated by a blank spaces. Example is shown below.

```
for (Rect rect : faceDetections.toArray()){  
  
}
```

3.3.1.5 Implementation comments

Here it has used comments delimited by `/*...*/` or `//`.

- Block Comments

Mostly block comments are used to provide a detailed description of methods, files, algorithms and data structures. Block comments are used at the beginning of each method and it has described about the function in order to enhance the readability of the user. Example is shown below.

```
/*
```

Detection of eyes with the use of Haarcascade file called haarcascade_eye.xml file
and call detect method to detect eyes

```
*/
```

- Commenting codes

The `//` comment delimiter is used in some instances to comment a single line of code so that user can understand the code clearly. Example is shown below.

```
//loading the Haarcascade file
```

3.3.1.6 Standards of statements

Compound statements which contains list of statements enclosed in braces are intended one more level than the compound statement. The opening brace is at the end of the line that begins the compound statement and the closing brace is in the beginning the line and be intended to the beginning of the compound statement. Braces should be used around all statements, even single statements, when they are part of a control structure, such as an if-else or for statement.

- Return Statements

A return statement with a value is not used parenthesis. Example is shown below.

```
return currentFrame;
```

- if statements

The “if” keyword and conditional expression is placed on the same line. The if statements are always use {} braces in order to improve the readability of the user. Example is shown below.

```
if(count==1 || count==2){

    Mat image_roi = new Mat(image,rect_Crop);

    Imgcodecs.imwrite(path+fi_name +count+"_Detected_Eye.jpg",image_roi);

    CroppedEyesName[count-1]=path+fi_name +count+"_Detected_Eye.jpg";

    count++;

}
```

- For statements

The keyword ‘for’ and the parenthesis is separated by a space and the expressions in a “for” statement should be separated by blank space. The statement block is placed on the next line. The closing curly brace starts in a new line, indented to match its for loop statement. Example is shown below.

```
for (int i = 0; i < total_Eyes; i++){

    CvRect r = new CvRect(cvGetSeqElem(sign, i));

    cvRectangle ( src, cvPoint(r.x(), r.y()), cvPoint(r.width() + r.x(),
    r.height() + r.y()), CvScalar.RED, 5, CV_AA, 0);

}
```

- Switch statements

The ‘switch’ construct should use the same layout format as the ‘if’ construct. The ‘switch’ keyword should appear on its own line, immediately followed by its test expression. The keyword ‘switch’ and the parenthesis should be separated by a space. The statement block should be placed on the next line. The closing curly brace starts in a new line, indented to match its corresponding opening statement. Every time a case falls through, i.e. does not include a break statement, a comment should be added where the break statement would normally be. Every switch statement must include a default case. A break in the default case is redundant, but it prevents error if another case is added later and therefore may be added. All missing switch case break statements should be correct and marked with a comment. Example is shown below.

```
public void onMessage(final int messageID, final Object message) {

    switch (messageID) {
```

```

case MessageHUB.MEASUREMENT_STEP:
    updateUI((MeasurementStepMessage) message);
    break;
case MessageHUB.DONE_CALIBRATION:
    _calibrateButton.setBackgroundResource(R.drawable.green_button);
    break;
default:
    break;
}
}

```

- try-catch Statements

In the try/catch construct the 'try' keyword is followed by the open brace in its own line. This is followed by the statement body and the close brace on its own line. This may follow any number of 'catch' phrases - consisting of the 'catch' keyword and the exception expression on its own line with the 'catch' body; followed by the close brace on its own line. Example is shown below.

```

try {
    newfile.createNewFile();
} catch (IOException e){
//body
}

```

3.3.1.7 Naming Conventions

Naming conventions make programs more understandable by making them more easier to read. Following conventions are followed by Eye Care developers while naming a class or a member.

Eye Care has used English descriptors that accurately describes the variable, method and class. Example is shown below.

- Variable name

```
public String oriImgSvNameWithPath;
```

- method name

```
CropEyes();
```

- Eye Care has used mixed case to make names readable with lowercase letters in general capitalizing the first letter of class names and interface names.

```
public class PreProcessing {
```

```
    //body
```

```
}
```

- Names of the classes or methods does not consist of very long or similar names.

```
public void CropEyes(){
```

```
    //body
```

```
}
```

```
public void EyeDetection(String imgName){

    //body

}
```

3.3.1.8 Standards for Methods

- Naming methods

Methods has named using a full English description, using mixed case with the first letter of any non-initial word capitalized. First word of method name is strong active verb. Example is shown below.

```
public void irisDetectionAndCropping(){

    //body

}
```

- Getters

Getters are the methods which are returning value of a field. ‘get’ is prefixed to the name of the field. Example is shown below.

```
getdistanceto Face();
```

- Setters

Setters which are also known as mutators are the methods which modify the values of a field. In set methods the word ‘set’ is prefixed to the name of the field. Example is shown below.

```
setContentView(R.layout.activity__self_cam);
```

- Method visibility

Methods which does not have to be private have defined as public. Member is only visible within the class, not from any other class is defined as private. Example is shown below.

```
private void resetCam() {
    _mySurfaceView.reset();
    _cam.stopPreview();
    _cam.setPreviewCallback(null);
    _cam.release();
}

public void EyeDetection(String imgName){

    String filePath="C:\\Users\\Mango\\Desktop\\re\\

    +ImageUploadWebApp\\UploadedImages\\";

    String XML_FILE =
"C:\\Users\\Mango\\Desktop\\re\\ImageUploadWebApp\\

    +CascadeFiles\\haarcascade_eye.xml";

    IplImage img = cvLoadImage(filePath+imgName);

    detect(img,imgName);

}
```

3.3.1.9 Naming Convention standards

- Naming variables

Variables are simple and they are very meaningful. Example is shown below.

```
public String fi_name;
```

```
public String oriImgSvNameWithPath;
```

- Naming components

Names of the components are meaningful with post fixed by the component type. This makes it easy to identify the purpose of the component as well as its type, making it easier to find each component in a list. It has avoided Button1 and Button2 names as ids'. Example is shown below.

- ✓ calibrateButton
- ✓ resetButton
- ✓ Naming constants

Constants whose values are not change are implemented as static final fields of classes. They have represented with full English words, all in upper case with underscores between the words like FINAL_VAL. Example is shown below.

```
public static final int CALIBRATION_DISTANCE_A4_MM = 294;
```

```
public static final int CALIBRATION_MEASUREMENTS = 10;
```

```
public static final int AVERAGE_THREASHHOLD = 5;
```

- Naming local variables

Local variables are named with the use of full English descriptors with the first letter of any non-initial word in lowercase. Example is shown below.


```
public String fi_name;
```

```
public String oriImgSvNameWithPath;
```

- Naming loop counters

Loop counters has used simple letters like i, k and j. Example is shown below.

```
for(int i=0;i<1;i++){  
  
}
```

- Variable Assignments

It has avoided assigning several variables to the same value in a single statement like value1=value2=value3=5;. Assignment operator is not used in a place where it can be easily confused with the equality operator. Example is shown below. Example is shown below.

```
int count=1;
```

3.3.1.10 Standards for classes

Class names have defined in simple with full English descriptor nouns, in mixed case starting with the first letter capital and others are in simple. Example is shown below.

```
public class PreProcessing {  
  
}
```

- Class visibility

Visibility of the classes are used as public. Example is shown below.

```
public class PreProcessing {

}
```

3.3.2 UI standards

3.3.2.1 Form elements

A form element is any item within a form used to collect data from a user. Most commonly it consists of a title, an input field, a description, and possibly a prefix, a suffix, or both.

3.3.2.2 Title

Title of the form element is simple and complete word which describes the form element. First character is capitalized and there are no ending punctuations.

3.3.2.3 Description

Questions are in sentence form and in simple English.

CONJUNCTIVITIES

1. Do you have itching?
☐ yes ☐ no
2. Are there any green or white discharge from your eye?
☐ yes ☐ no
3. Do you have tearing in your eyes?
☐ yes ☐ no
4. Do you feel Burning eyes?
☐ yes ☐ no
5. Do you have any Blurring in vision?
☐ yes ☐ no
6. Do you have a headache with the flu?
☐ yes ☐ no

Figure 3.7: Question form

Figure 3.7 shows the Questionnaires given to the users to answer and user needs to click submit button in order to get the results.

The screenshot displays a mobile application interface for a questionnaire. At the top, the status bar shows 'Mobitel', '0K/s', '100%' battery, and '2:04 pm'. The questionnaire consists of seven questions, each with two radio button options: 'yes' and 'no'. The questions are:

- 3. Do you have tearing in your eyes?
- 4. Do you feel Burning eyes?
- 5. Do you have any Blurring in vision?
- 6. Do you have a headache with the flu?
- 7. Do you have fainting?

At the bottom of the form is a blue button labeled 'Submit'. The bottom of the screen shows the standard Android navigation bar with back, home, and recent apps icons.

Figure 3.8: Questionnaire form

Figure 3.8 shows questionnaires given to the users to answer and user needs to click submit button in order to get the results.

3.3.2.4 Buttons

Buttons are designed in blue color with the white text.

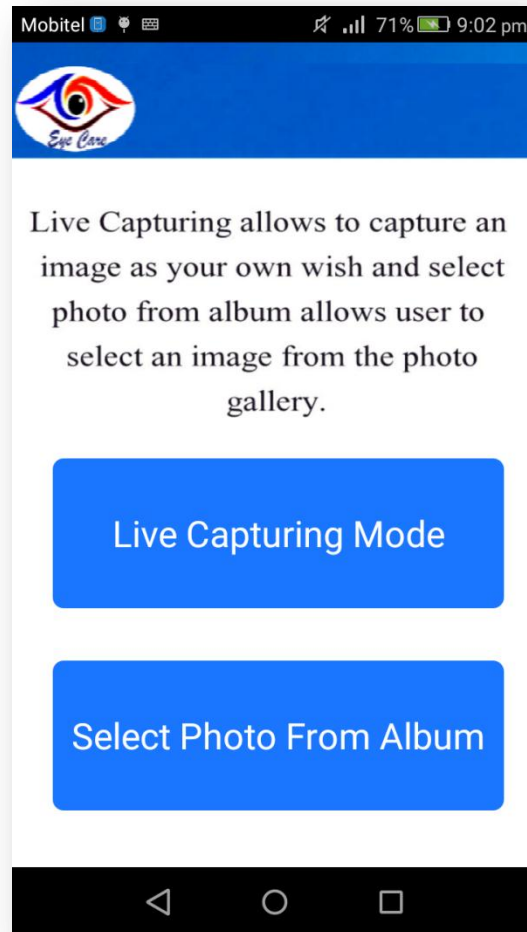


Figure 3.9: Option selection

Figure 3.9 shows how User can test the eye with the use of Test your eye button. User can click How to use eye care in order to see how Eye Care works and user can get a clear understanding about the “Eye Care” application by clicking About Eye Care button.

3.3.2.5 Interface text

Interface text has used neutral and polite tone. Interfaces does not include “we”, “sorry” and “please” keywords. It has avoided emotional verbs and adjectives.

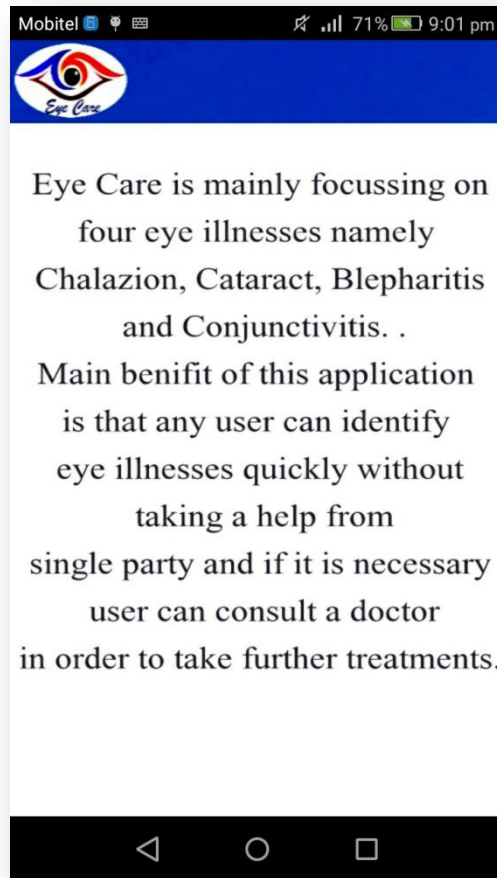


Figure 3.10: Image selection option

Figure 3.10 provides user to select an image from the album or to capture an image with the use of camera.

3.3.2.6 UI-elements

More detailed descriptions are displayed in the “How to use Eye Care” page.

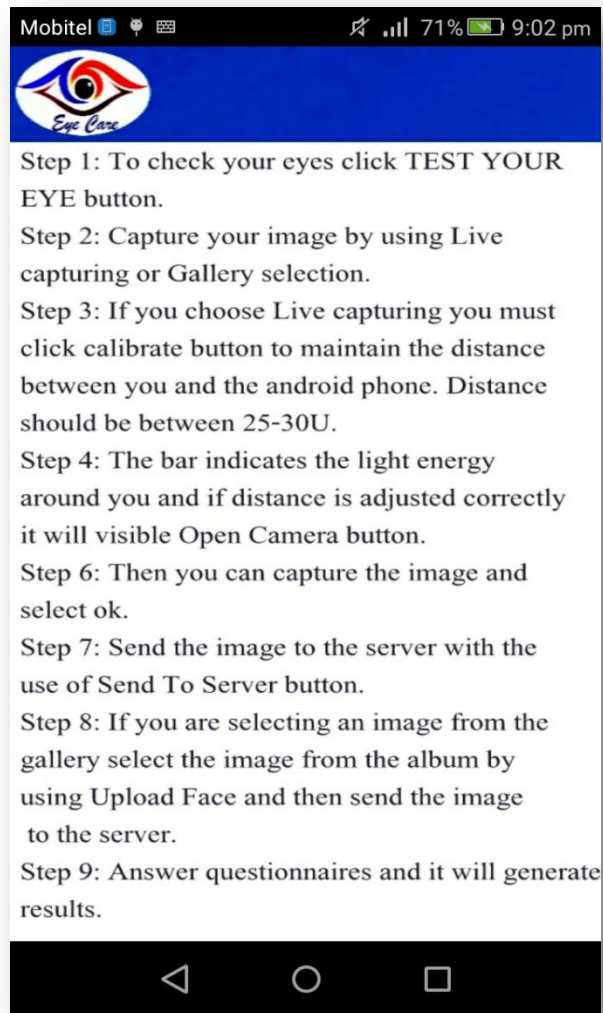


Figure 3.11: How you use Eye Care

Figure 3.11 interface helps the user to get a clear understanding on how to use the application

3.3.2.7 Wording

Each and every button of the Eye Care application has used simple and meaningful words to enhance the readability of the user.

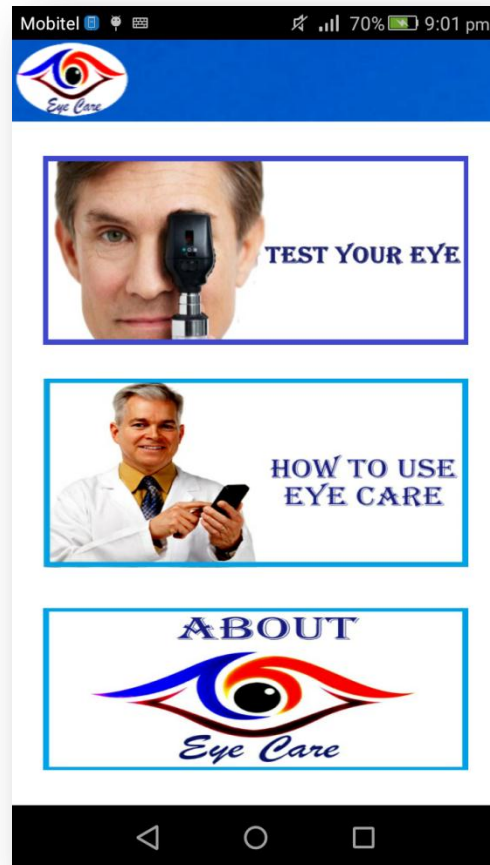


Figure 3.12: Loading user's image

Figure 3.12 shows how image is loaded from the gallery and user can upload the image to the server with the use of Upload Face button.

3.3.2.8 Capitalization

Page titles, headings and names of the buttons generally use “sentence case” where only the first letter is capitalized.

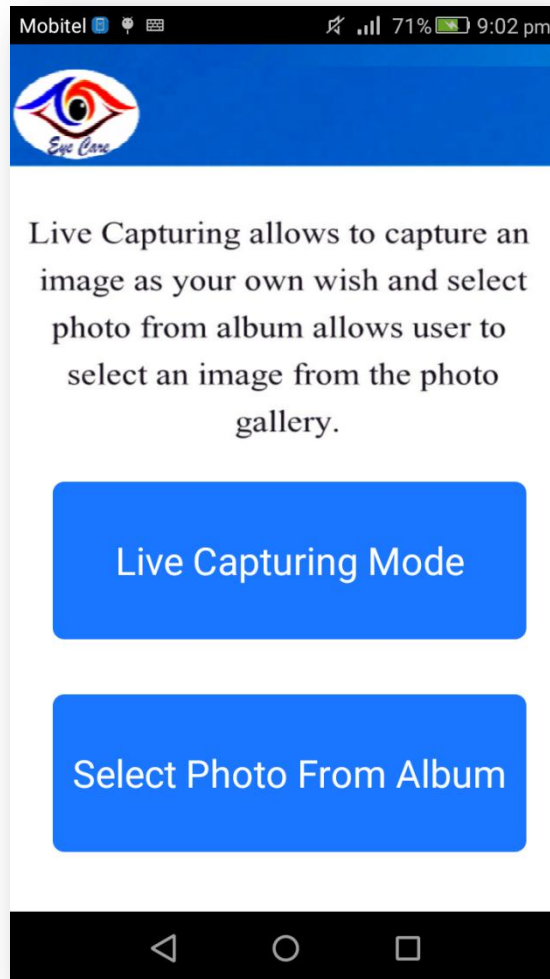


Figure 3.13: Select Option interface

Figure 3.13 shows how user can select any option according to the user preference.

3.3.2.9 Banners and background theme

Each and every page of the Eye Care use the below mentioned banner theme. It has used blue and white color theme so that any user can interact with the application very easily.



Figure 3.14: Eye Care Banner

Figure 3.14 depicts the banner of the Eye Care application which displays in all the interfaces.

3.4 Nonfunctional requirements

3.4.1 Reliability

Software Reliability is the probability of failure-free software operation for a specified period of time in a specified environment [62]. The “Eye Care” application should not be failed to give incorrect output since it is considered as a medical application. Therefore efficient and appropriate Pre-Processing techniques must be used to increase the accuracy of the “Eye Care” application. As well as that hardware with good quality should be used and they should be tested and maintained regularly to avoid hardware failures and the entire system should be thoroughly tested before releasing to find and fix bugs. Similarly a beta version of the system can be released to detect the bugs and other obvious requirements and change requests of the users provide the expected output with a higher level of accuracy.

3.4.2 Availability

“Availability defines the proportion of time that the system is functional and working. It can be measured as a percentage of the total system downtime over a predefined period. Availability will be affected by system errors, infrastructure problems, malicious attacks, and system load.”[63] Users of the “Eye Care” application may use the application every day to check their eye illness status. Therefore in order to achieve a higher level of user satisfaction the system’s availability should be around 99%. In other words the system should be available 365 days per year, 744 hours per month and 168 hours per week. To guarantee this level of availability a proper backup schedule to take regular backups of databases should be maintained, hence the unavailability occurs due to server failures or database crashes can be avoided. The system is built to communicate using ant the type of data communication which is available (4G, 3G, or Wi-Fi).

3.4.3 Security

“Security is the capability of a system to reduce the chance of malicious or accidental actions outside of the designed usage affecting the system, and prevent disclosure or loss of information. Improving security can also increase the reliability of the system by reducing the chances of an attack succeeding and impairing system operation.” [64] Since private details of users and private images are passed over the network it’s really important to have a secure connection.

3.4.4 Maintainability

Maintainability is the ability of the system to undergo changes with a degree of ease.” [65]. With the release of the “Eye Care” application many change requests may appear in order to fix errors, or to meet new business requirements. When adding or changing the application’s functionality in order to fulfill the change requests they may impact components, services, features and interfaces of the “Eye Care” application. Maintainability can also affect the time it takes to restore the system to its operational status following a failure or removal from operation for an upgrade. Improving system maintainability can increase availability and reduce the effects of run-time defects. The “Eye Care” system can achieve a high availability since it follows object oriented concepts. The “Eye Care” system is designed as well-defined layers, or areas of concern, that clearly separate the system’s UI, business processes, and data access functionality. Maintainability can be increased further by reducing the dependencies between the modules. Meantime the source code will also use advance coding standards with adequate inline and header comments and methodical documentation for further development.

3.5 Other Requirements

3.5.1 User-friendliness

The graphical user interfaces should reflect attraction and enthusiasm upon users to perform the activities using the system easily. The application is consisting of unique color theme and it is very easy to access by any user without taking any help from third party.

3.5.2 Simplicity

All the user interfaces of the system should be designed in a simple and straight forward manner in order to can use this application by everyone; they should be able to operate with the system without confusion. Eye Care application has used simple English so that any user can understand very easily. Since there is an easy navigation system user can be able to operate the application very easily.

3.6 Commercialization of the product

3.6.1 Target Market

People those who are suffering from eye illnesses can use this application in order to check their eye illnesses. But there is no restriction to use this application only by eye illness patients. Those who are willing to use the Eye Care application simply they can use the application in order to check their eye health status.

3.6.2 Marketing Mix

3.6.2.1 Product

Eye Care is a mobile based application where it is mainly focusing on four eye illnesses namely Chalazion, Cataract, Blepharitis and Conjunctivitis. At present still there are no commercially available mobile applications which recognize these four eye illnesses. The uniqueness of this product is that the illnesses are identifying with a combination of two main techniques called image processing and knowledge base. Main advantage of this application is that any user can use the application without taking help from others since the application is more user friendly when comparing to the existing applications that are available in the market. "Eye Care" application mainly targeting eye patients but it can be used by anyone who use android phones. Therefore this system will be very useful for the people in order to check whether he/she is suffering from the above eye illnesses.

- Uniqueness of Eye Care

Eye Care is a mobile based application where it detects four prominent eye illnesses. It provides the details of eye consultants and home remedies to minimize the illness condition. Illness is identified with the help of two main techniques called image processing and semantic web in order to maintain a good accuracy level.

- Benefits of these features

Main benefit of this Eye Care application is that any user can clearly identify the eye illness without taking any help from a third party. User can contact eye specialists very easily those who are related to the illnesses and if they need they can check their eye illness without going to the hospital. Eye Care provides user with simple home remedies in order to minimize the illnesses.

- Product positioned against the competition

Eye Care application detects main four eye illnesses namely Chalazion, Conjunctivitis, Cataract and Blepharitis. Still there are no applications available in the market which detects the above eye illnesses. Therefore this application still has the chance to compete with the existing applications in the app market.

- Difference between Eye Care and Existing applications

Major difference between Eye Care and existing application is that Eye Care identifies four common eye illnesses of a patient with the use of two advance techniques called image processing and Semantic web where existing applications have not focused on detecting above eye illnesses so far.

- History of the product in the market place

Since there are no applications available to detect common eye illnesses named Chalazion, Blepharitis, Conjunctivitis and Chalazion Eye Care will become a brand new product to the app market.

3.6.2.2 Price

In analyzing the results of the market survey, it was found that the eye illness patient population constituted 90% of the market. Also, this patient population was found to be much more price inelastic, willing, on average, to pay \$30-\$40.

- Business model used by Eye Care

A business model needs to get an idea about how to gain revenue. This is the most challenging aspect when launching a mobile application. Eye Care used freemium business model by which a product or service is provided free of charge but money is charger for special features, functionalities and virtual goods.

Eye Care mobile application can be downloaded by Google Play Store and Apple Store. It is available free. Here the application provides to detect two illnesses free and in order to detect the other two eye illnesses user needs to pay \$100. Eye Care application facilitates patients to view eye consultants and the home remedies which he/she can take in order to minimize the eye illness. In order to view the set of eye consultants user needs to pay \$5 and to view home remedies related to the illness he/she needs to pay extra \$7.

3.6.2.3 Place

Eye Care application can be downloaded from Google Play store which is Google's official pre-installed app on Android-trademarked devices which provides access to content on the Google play store. It allows user to download music, books, magazines and other mobile applications very easily.

3.6.2.4 Promotion

Main objective of this promotional strategies are to make customers aware about the availability of Eye Care application and to convert them to use Eye Care. Marketing strategies can be divided in to three main categories namely Direct Sales, Direct Marketing and Mass Marketing. In direct sales it is directly reaching to the customers via personal contacts, telephones and email.

In direct marketing it provides communication to the customers via advertising via websites. In mass marketing it deliver the message to everyone who wishes to listen to the news with the help of television, internet and social Medias. In order to promote the product Eye Care application development team used mass marketing where anyone can hear the get an idea about the product very easily.

According to the reports and records from eye hospitals it can be concluded that 90% people are suffering from eye illnesses. Since these people are not current customers of Eye Care product it has no relationship with users. Currently the development team does not have any contact details it is trying to market the product especially through internet. Therefore it is mainly focusing on social media publications in order to promote the application.

- Building a microsite

Microsite has the power to highlight Eye Care product. It helps to launch Eye Care innovative product in a way that full websites or more traditional marketing cannot. Microsite contains a single or a small collection of pages which conveys the relevant information in a creative way.

- Building a teaser website

A teaser can go a long way to ensuring a successful launch. Raising awareness of Eye Care product long before it ships makes sense: people are more likely to interact with Eye Care app if they're somewhat familiar with it, so letting them hear about it as soon (and as much) as possible helps. Ideally Eye Care development team would create a climate of anticipation, and make people look forward to the release date. And also it can collect email addresses of people who are willing to download the Eye Care application.

- Creating a blog

Eye Care has created a blog which is attached to microsite where it contains interesting content that draw people to the website.

- Create social media accounts

In order to promote Eye Care application Eye Care development team will create Twitter, Facebook, Google+ and LinkedIn profiles so that it can promote the Eye Care app several times a week.

- Creation of product video

Eye Care will create a product video in order attract the customers to download this application via Google app store. In order to retrieve the reviews of the users Eye Care will create a review website so that it can get a clear understanding about how to develop Eye Care further in order to satisfy the customers.

- Eye Care will contact bloggers who write on niches that relate to Eye Care application.

Eye Care will contact with bloggers those who are writing on niches which are related to Eye Care application. So that Eye Care can be promote very easily.

- Create a Facebook group

Eye Care team will develop a Facebook group. Here the group will consist of similar interest people related to eye applications so that they can meet and socialize regularly.

- Contact admins of Facebook pages

Eye Care team will contact admins of Facebook pages which consists of about 100k+ likes so that it will help to promote the Eye Care app to a wider range of people. Contacting Facebook pages should be related to medical field.

Marketing strategy can be divided in to two main parts namely pre-launch and post-launch strategies. In prelaunch strategy it used to create awareness and visibility with the goal of acquiring the interest and building the curiosity of as many users as possible. Post-launch strategy is used to keep in touch with existing customers.

3.6.2.5 Ten days marketing plan to launch Eye Care

- Before launch (6 days before)

Creates a landing page for Eye Care app to figure out the right messaging to Eye Care app. To create the landing page team uses <http://unbounce.com> where it has a free theme for mobile apps and it can create Eye Care landing page very quickly. In landing page it is going to mention that Eye Care will launch the application soon and if users' wants an invite they can provide their email addresses.

- Five days before launch the app

After landing page is created then it is the time to send relevant traffic to Eye Care landing page. These will help to validate idea as well as to get an idea how many are wishing to download Eye Care application once it launches. In order to send traffic jam to landing page of Eye Care it creates Facebook ads and drive traffic to the landing page.

- Four days before launch

Good press coverage for the Eye Care application helps to get a huge number of customers on the launch day. First team will find reporters who cover app launches. Then the company can reach out these reporters and tell them about the application.

- Three days before launch

Then team will create app store description which contains headline, features and a few testimonials and get started with app store optimization.

- Two days before launch

As it is getting closer to the launch day, it needs to ensure that the analytics tool which team is using to track this application is properly setup. It must verify that has created events for the major checkpoints in Eye Care app and also setup funnels to track progress of users in Eye Care app.

- One day before launch

It is time to create an Instagram in order to get the customers attention towards the application.

- Launch Day

Make sure that all the reporters who were supposed to write about Eye Care application have actually done that and also send out emails to the relevant people who provided their email addresses informing them about the launch app.

- One day after Launch

After seeing the initial set of downloads then use Facebook to create some advertisements about the application so that it can be useful get the popularity towards the application.

- Two days after launch

Launch Instagram sponsored posts through the partnerships which you set up before the launch.

- Three days After Launch

A use giveaway to get thousands of downloads.

- Finally

Ratings and Reviews will encourage Eye Care customers to review and rate our application. If the rating is high, the more chances of it being downloaded from the App Store.

3.6.3 Maintenance

Software maintenance is important for many important reasons. The main reason for maintaining the software is that the software and hardware is constantly changed. Therefore it is very important to keep applications synchronized with each other. When new technologies arrive, it is a must to modify the application's interfaces in order to incorporate new hardware or software. Application features must be change in order to satisfy the customers of the Eye Care application since their requirements changes time to time.

3.6.4 How to contact with customers

Before launching the application Eye Care development team will try to build up its own organization. Following are the details of the organization which is going to form by the development team so that the customers can contact whenever they meets with any concern.

3.6.4.1 Business Details

Table 3.3: Business Details

Business name	Enterprise Solutions for Mobile Applications and Cloud Computing
Owners	Oshani Wickramasingha, Angel Gunarathne, Chanaki Sugathapala, Anupama Soysa
Trading name	ESMACC
Established	November 5, 2016
Business Formation	Partnership
Business Operation	Full time venture

	Monday to Friday 8:00 am to 9:00 pm
Business Description	Developing medical mobile based applications
Location	No 32, Havelock Road, Colombo 06
Primary line of Business	Eye Care mobile based application

3.6.4.2 Contact details

Table 3.4: Contact details

Contact name	Oshani Tharanga
Landline	+94 011 2341 203
Mobile	+94 071 5788 696
Email	Esmaccsoft@gmail.com
Fax	+94 81 385 707
Physical address	No 32, Havelock Road, Colombo 06
Postal address	No 32, Havelock Road, Colombo 06

3.6.4.3 Online/ Social Media

Table 3.5: Online/social media

Website	www.esmac.ccom
LinkedIn	https://lk.linkedin.com/in/esmacc-devoopers-48242899
Facebook	https://web.facebook.com/esmacc.2016/?ref=aymt_homepage_panel
YouTube	https://www.youtube.com/watch?v=esmacc
Blog	http://esmacc_pvt.com/

3.6.4.4 Advisors

Table 3.6: Advisors

Name	Role	Address
Dr. I. L. M. Rifaz	Director of National Eye Hospital, Colombo	National Eye Hospital, Colombo
Dr. Prasad Colabage	Senior eye surgeon	National Eye Hospital, Colombo

4 CONCLUSION

At present people are running out of their time and they do not take care about their health status. As a result people used to take home remedies. Sometimes they try to contact people from the internet in order to get simple treatments. This method can be too dangerous since they are lack in medical knowledge.

“Eye Care” is a mobile android application that can be used among any user in order to check four eye illnesses namely Chalazion, Blepharitis, Cataract and Conjunctivitis. It detects the illness by using Image Processing technique and then by using semantic web technique. In order to get more accurate results user should answer some questions, that will be provided by the application and finally “Eye Care” is going to generate results with some home precautions as well as a list of eye specialists. Then person can get an understanding about the illness and if it is an emergency user can consult a doctor immediately.

Figure 4.1 and figure 4.2 were taken by the national eye hospital Colombo. These images depict the percentage of surgeries performed in 2014 and the number of surgeries done from 2009 to 2014.

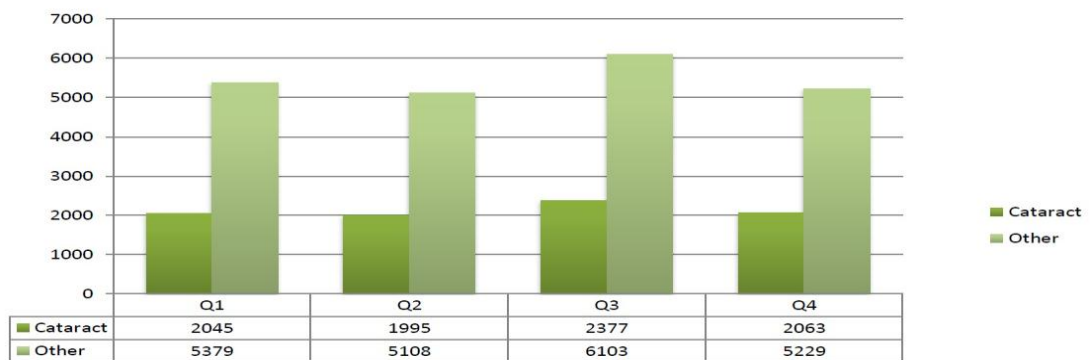


Figure 4.1: Surgeries performed in 2014

Source: The above figure is extracted from annual report 2014 which has published by National Eye Hospital [67].

Figure 4.1 shows the percentage of surgeries done in 2014 at National Eye Hospital.

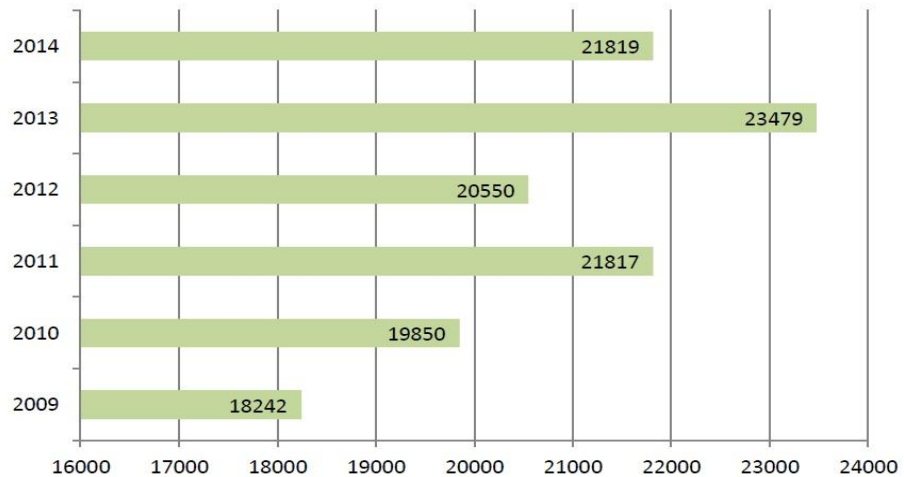


Figure 4.2: Surgeries done from 2009 to 2014

Source: The above figure is extracted from annual report 2014 which has published by National Eye Hospital [68].

Figure 4.2 shows the number of surgeries done from 2009 to 2014 in National Eye Hospital.

There is less number of researches based on image processing to identify eyes and to crop eyes. And still there is no Haar Cascade file is written in order to detect iris and to crop iris. Preprocessing and segmentation module consists of two major parts namely android application and server.

4.1 Image Preprocessing Findings

- Configuring android application in order to capture user's image.
- Accessing smart phone memory and retrieving data from memory.
- Exchanging data and images between android application and java server
- Calibrating the distance between the user and the android phone
- Measuring the light source
- Creating a Haar cascade file to detect iris

4.2 Research Contribution To The Field

Any user can use the “Eye Care” application in order to check whether he/she is suffering from main four eye illnesses namely Cataract, Blepharitis, Conjunctivitis and Chalazion. Main advantage of “Eye Care” application is that the user can use the application without taking any single help from a third party. If user needs to get some home precautions “Eye Care” application will give some home remedies in order to minimize the illness up to some level. “Eye Care” application will display a list of eye specialist where user can channel them very quickly if he/she is in an emergency condition.

4.3 Research Limitations

By image preprocessing and segmentation module it detects and crop irises of the provided image. In iris detection and iris cropping it detects and crop irises accurately. But when images are saving in the folder they save with some garbage images. Here the system cannot identify the iris directly. When user checks the image folder he/she can view the irises photos clearly. So this is one of the limitations of this image preprocessing and segmentation module.

When user enters an image with poor quality it will minimize the accuracy level of the application and may fall down the server as well. This application is only support for

android mobile phones and it is not supporting for other smart phones. When a user input a group photo it will stop the application since the application detect only one face at a time. This application only focusing on four eye illnesses by using image processing and knowledge base. These are the limitations that can be found in Eye Care mobile application. A research paper which is shown in Appendix K shows the limitations of the Eye Care project.

4.4 Recommendations For Future Research

In order to overcome the above mentioned problem developer can train a Haar cascade file where it needs about 60 hours to create the Haar cascade file. User must enter about 2000 images of positive images and 5000 images of negative images in order to create a better Haar cascade file.

Here quality of the picture is an issue. If user captures an image from a poor quality camera it is hard to identify iris and eyes in the server application. Therefore in order to minimize the above problem developer must build new Haar cascade files or algorithms which supports for the low quality cameras.

If the image contains a group photo or else other images server is falling down since the application is detecting only two eyes in a human image. If someone wishes to improve Eye Care application he/she can find a way to detect eyes in a group photo.

This application is only focusing on four eye illnesses. Therefore a person can improve the system which can detect several illnesses. Eye Care application supports only for android phones. Therefore a developer can build Eye Care application which will support for several platforms.

REFERENCES

- [1] WHO, "Visual impairment and blindness," in World Health Organization, World Health Organization, 2016. [Online]. Available: <http://www.who.int/mediacentre/factsheets/fs282/en/>. [Accessed: Jul. 25, 2016].
- [2] National Eye Institute, "National eye institute," in National eye institute, 2015. [Online]. Available: https://nei.nih.gov/health/cataract/cataract_facts. [Accessed: Jul. 25, 2016].
- [3] National Eye Institute, "Facts about Blepharitis," in National Eye Institute, 2009. [Online]. Available: <https://nei.nih.gov/health/blepharitis/blepharitis>. [Accessed: Jul. 25, 2016].
- [4] M. MacGill and H. Webberley, "Blepharitis: Causes, symptoms and treatment," Medical News Today, 2016. [Online]. Available: <http://www.medicalnewstoday.com/articles/185155.php>. [Accessed: Jul. 25, 2016].
- [5] WebMD, "What is Conjunctivitis, or Pinkeye?," in WebMD, WebMD, 2005. [Online]. Available: <http://www.webmd.com/eye-health/eye-health-conjunctivitis>. [Accessed: Jul. 25, 2016].
- [6] T. Shields and P. Sloane, "A comparison of eye problems in primary care and ophthalmology practices," Family medicine., vol. 23, no. 7, pp. 544–6, Sep. 1991. [Online]. Available: <http://www.ncbi.nlm.nih.gov/pubmed/1936738>. [Accessed: Jul. 25, 2016].
- [7] H. Kaufman, "Adenovirus advances: New diagnostic and therapeutic options," Current opinion in ophthalmology., vol. 22, no. 4, pp. 290–3, May 2011. [Online]. Available: <http://www.ncbi.nlm.nih.gov/pubmed/21537185>. Accessed: Jul. 25, 2016.

- [8] 2016 H. Media et al., "Chalazion," in healthline, Healthline, 2005. [Online]. Available: <http://www.healthline.com/health/chalazion#Overview1>. [Accessed: Jul. 25, 2016].
- [9] Engineers Garage, "Comments," in Engineers Garage, 2014. [Online]. Available: <http://www.engineersgarage.com/articles/image-processing-tutorial-applications>. [Accessed: Jul. 25, 2016].
- [10] "Image segmentation,". [Online]. Available: <https://www.cs.auckland.ac.nz/courses/compsci773s1c/lectures/ImageProcessing-html/topic3.htm>. [Accessed: Jul. 25, 2016].
- [11] "Image processing," in Wikipedia, Wikimedia Foundation, 2016. [Online]. Available: https://en.wikipedia.org/wiki/Image_processing. [Accessed: Jul. 25, 2016].
- [12] D. Health and Fitness, "Eye test - Android Apps on Google play," in GooglePlay, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.b2creativedesigns.eyetest&hl=en>. [Accessed: Jan. 12, 2016].
- [13] healthcare mobile 4 Health and Fitness, "Eye exercises - eye care plus – Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.eyexamtest.eyecareplus>. [Accessed: Jan. 12, 2016].
- [14] R. L. Health and Fitness, "Vision test 2.0 - Android Apps on Google play," in Google Play, 2015. [Online]. Available: <https://play.google.com/store/apps/details?id=com.threesidedcube.visiondroid&hl=en>. [Accessed: Jan. 12, 2016].
- [15] G. EyeVentures, "EyeXam on the App store," App Store, 2016. [Online]. Available: <https://itunes.apple.com/en/app/eyexam/id357290178?mt=8>. [Accessed: Jan. 12, 2016].

- [16] healthcare mobile 4 Health and Fitness, "Visual Acuity test - Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.eyexamtest.acuity&hl=en>. Accessed: Jan. 12, 2016.
- [17] G. EyeVentures, "EyeXam on the App store," App Store, 2016. [Online]. Available: <https://itunes.apple.com/en/app/eyexam/id357290178?mt=8>. [Accessed: Jan. 22, 2016].
- [18] healthcare mobile 4 Health and Fitness, "Eye exercises - eye care plus - Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.eyexamtest.eyecareplus&hl=en>. [Accessed: Jan. 12, 2016].
- [19] O. S. Medical, "Amsler grid - Android Apps on Google play," in Google Play, 2014. [Online]. Available: <https://play.google.com/store/apps/details?id=com.ossibussoftware.amsler&hl=en>. [Accessed: Jan. 2, 2016].
- [20] V. F. Tools, "Color blind pal - Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.colorblindpal.colorblindpal&hl=en>. Accessed: Jan. 3, 2016.
- [21] WebMD, "What is Conjunctivitis, or Pinkeye?," in WebMD, WebMD, 2005. [Online]. Available: <http://www.webmd.com/eye-health/eye-health-conjunctivitis>. [Accessed: Jul. 25, 2016].
- [22] healthcare mobile 4 Health and Fitness, "Visual Acuity test - Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.eyexamtest.acuity&hl=en>. [Accessed: Jan. 12, 2016].

- [23] A. U. Patwari, M. D. Arif, N. A. Chowdhury, A. Arefin, and I. Imam, "Detection, Categorization, and assessment of eye cataracts using digital image processing," in ResearchGate, ResearchGate, 2011. [Online]. Available: https://www.researchgate.net/publication/216690188_Detection_Categorization_and_Assessment_of_Eye_Cataracts_Using_Digital_Image_Processing. [Accessed: Jan. 12, 2016].
- [24] "Image processing," in Wikipedia, Wikimedia Foundation, 2016. [Online]. Available: https://en.wikipedia.org/wiki/Image_processing. [Accessed: Jan. 13, 2016].
- [25] "Image segmentation," in Wikipedia, Wikimedia Foundation, 2016. [Online]. Available: https://en.wikipedia.org/wiki/Image_segmentation#Histogram-based_methods. [Accessed: Jan. 2, 2016].
- [26] D. Anandhi, M. S. Josephine, and V. Jeyabalaraja, "Iris Segmentation and Detection System for Human Recognition Using Canny Detection Algorithm," IJAR: IJAR, 2015, p. 8. [Online]. Available: https://scholar.google.com/scholar?q=Iris+Segmentation+and+Detection+System+for+Human+Recognition+Using+Canny+Detection+Algorithm&hl=en&as_sdt=0&as_vis=1&oi=scholar&sa=X&ved=0ahUKEwjfs7OAs7LLAhWIC5oKHU1qA8EQgQMIGDAA. [Accessed: Jan. 24, 2016].
- [27] Ashish Kare.class lecture,Topic: "Image Pre-Processing", school of Electrical and Computer Engineering,Dec.6,2012.
- [28] Vincent Luboz. Class Lecture, Topic: "Image Segmentation and Pre - processing." Department of Surgery and Cancer,Imperial College,London.
- [29] T. Rajpathak, R. Kumar, and E. Schwartz, "Eye Detection Using Morphological and Color Image Processing," in Florida Conference on Recent Advances in Robotics,

2009, p. 1-6. [Online]. Available: http://www.mil.ufl.edu/publications/fcrar09/eye_detection_tanmay_rajpathak_fcrar_09.pdf. [Accessed: Aug. 2, 2016].

[30] e. <>, "Histogram equalization — OpenCV 2.4.13.0 documentation," in OpenCv, 2011. [Online]. Available: http://docs.opencv.org/2.4/doc/tutorials/imgproc/histograms/histogram_equalization/histogram_equalization.html. [Accessed: Feb. 12, 2016].

[31] OpenCv, "OpenCV: Face detection using Haar cascades," in OpenCv. [Online]. Available: http://docs.opencv.org/3.1.0/d7/d8b/tutorial_py_face_detection.html#gsc.tab=0. [Accessed: Feb. 12, 2016].

[32] , "Jaeger chart," in Wikipedia, Wikimedia Foundation, 2016. [Online]. Available: https://en.wikipedia.org/wiki/Jaeger_chart. Accessed: Jan. 3, 2016.

[33] WHO, "Visual impairment and blindness," in World Health Organization, World Health Organization, 2016. [Online]. Available: <http://www.who.int/mediacentre/factsheets/fs282/en/>. [Accessed: Jul. 27, 2016].

[34] National Eye Institute, "Eye Disease Statistics," Mar. 2014. [Online]. Available: https://nei.nih.gov/sites/default/files/nei-pdfs/NEI_Eye_Disease_Statistics_Factsheet_2014_V10.pdf. [Accessed: Jul. 27, 2016].

[35] M. MacGill and H. Webberley, "Blepharitis: Causes, symptoms and treatment," Medical News Today, 2016. [Online]. Available: <http://www.medicalnewstoday.com/articles/185155.php>. [Accessed: Jul. 27, 2016].

- [36] A. A. Azari and N. P. Barney, "Conjunctivitis," *Conjunctivitis*, vol. 310, no. 16, Oct. 2013. [Online]. Available: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4049531/>. [Accessed: Jul. 27, 2016].
- [37] B. Chou, "Chalazion," in *allaboutvision*, All About Vision, 2016. [Online]. Available: <http://www.allaboutvision.com/conditions/chalazion.htm>. [Accessed: Jul. 31, 2016].
- [38] "SDLC - quick guide," in *tutorialspoint*, www.tutorialspoint.com, 2016. [Online]. Available: http://www.tutorialspoint.com/sdlc/sdlc_quick_guide.htm. [Accessed: Jul. 22, 2016].
- [39] "Waterfall model," in *Wikipedia*, Wikimedia Foundation, 2016. [Online]. Available: https://en.wikipedia.org/wiki/Waterfall_model. [Accessed: Jul. 21, 2016].
- [40] "What is Iterative model- advantages, disadvantages and when to use it?," in *ISTQB Exam Certification*. [Online]. Available: <http://istqbexamcertification.com/what-is-iterative-model-advantages-disadvantages-and-when-to-use-it/>. [Accessed: Jul. 21, 2016].
- [41] "What is Spiral model- advantages, disadvantages and when to use it?," in *ISTQB Exam Certification*. [Online]. Available: <http://istqbexamcertification.com/what-is-spiral-model-advantages-disadvantages-and-when-to-use-it/>. [Accessed: Jul. 21, 2016].
- [42] "SDLC - big bang model," in *tutorialspoint*, www.tutorialspoint.com, 2016. [Online]. Available: http://www.tutorialspoint.com/sdlc/sdlc_bigbang_model.htm. [Accessed: Jul. 21, 2016].
- [43] "SDLC - agile model," in *tutorialspoint*, www.tutorialspoint.com, 2016. [Online]. Available: http://www.tutorialspoint.com/sdlc/sdlc_agile_model.htm. Accessed: Jan. 4, 2016.

[44] "SDLC - RAD model," in tutorialspoint, www.tutorialspoint.com, 2016. [Online]. Available: http://www.tutorialspoint.com/sdlc/sdlc_rad_model.htm. Accessed: Jan. 5, 2016.

[45] Posted and M. Rouse, "What is Prototyping model? - definition from WhatIs.com," in tutorialspoint, SearchCIO, 2005. [Online]. Available: <http://searchcio.techtarget.com/definition/Prototyping-Model>. Accessed: Jan. 4, 2016.

[46] melsatar, "Software development life cycle models and methodologies," in Software Engineering Practices, Mohamed Sami, 2012. [Online]. Available: <https://melsatar.wordpress.com/2012/03/15/software-development-life-cycle-models-and-methodologies/>. [Accessed: Jul. 21, 2016].

[47] B. Chou, "What causes eyelid inflammation (blepharitis)?," All About Vision. [Online]. Available: <http://www.allaboutvision.com/conditions/blepharitis.htm>. Accessed: Jan. 10, 2016.

[49] G. Bailey, "Hazy, blurry vision means you might have cataracts. Learn more in this complete guide to cataracts," All About Vision. [Online]. Available: <http://www.allaboutvision.com/conditions/cataracts.htm>. Accessed: Jan. 6, 2016.

[48] 2016 H. Media et al., "Chalazion," Healthline, 2005. [Online]. Available: <http://www.healthline.com/health/chalazion#Overview1>. Accessed: Jan. 6, 2016.

[50] "What is Conjunctivitis, or Pinkeye?," in WebMD, WebMD, 2005. [Online]. Available: <http://www.webmd.com/eye-health/eye-health-conjunctivitis>. Accessed: Jan. 10, 2016.

[51] G. Heiting, "Eye news: Visual flicker test may detect dementia more quickly and easily than MRIs and other tests," All About Vision. [Online]. Available: <http://www.allaboutvision.com/conditions/>. Accessed: Jan. 6, 2016.

- [52] 2016 H. Media et al., "See how the eye works," Healthline, 2013. [Online]. Available: <http://www.healthline.com/health/eye-health>. Accessed: Jan. 12, 2016.
- [53] Use case diagram Actor-VP Gallery, UML Modeling. [Online]. Available: <http://www.visual-paradigm.com/VPGallery/diagrams/UseCase.html>, [Access: May 2, 2013]
- [54] A. T. Project, "Apache Tomcat® - welcome!," 1999. [Online]. Available: <http://tomcat.apache.org/>. Accessed: Mar. 2, 2016.
- [55] "OpenCV," 2016. [Online]. Available: <http://opencv.org/>. Accessed: Jan. 9, 2016.
- [56] MvnRepository, "Android Async Http library," 2006. [Online]. Available: <https://mvnrepository.com/artifact/com.loopj.android/android-async-http/1.4.4>. Accessed: Jan. 9, 2016.
- [57] "Cascade classification — opencv v2.2 documentation," 2010. [Online]. Available: http://opencv.jp/opencv-2.2_org/c/objdetect_cascade_classification.html. Accessed: Feb. 3, 2016.
- [58] "Cascading classifiers," in Wikipedia, Wikimedia Foundation, 2016. [Online]. Available: https://en.wikipedia.org/wiki/Cascading_classifiers. Accessed: Mar. 3, 2016.
- [59] Theme and C. Powered, "Unit testing," in Software testing fundamentals, 2016. [Online]. Available: <http://softwaretestingfundamentals.com/unit-testing/>. Accessed: Aug. 23, 2016.
- [60] "What is user acceptance testing? Definition and meaning," BusinessDictionary.com, 2016. [Online]. Available: <http://www.businessdictionary.com/definition/user-acceptance-testing.html>. Accessed: Aug. 20, 2016.

- [61] Theme and C. Powered, "System testing," 2016. [Online]. Available: <http://softwaretestingfundamentals.com/system-testing/>. Accessed: Aug. 20, 2016.
- [62] J. Pan, "Software reliability," in *Software reliability*, 1999. [Online]. Available: https://users.ece.cmu.edu/~koopman/des_s99/sw_reliability/. Accessed: Oct. 24, 2016.
- [63] Microsoft, "Chapter 16: Quality attributes," 2016. [Online]. Available: <https://msdn.microsoft.com/en-us/library/ee658094.aspx>. Accessed: Oct. 24, 2016.
- [64] Microsoft, "Chapter 16: Quality attributes," 2016. [Online]. Available: <http://msdn.microsoft.com/en-us/library/ee658094.aspx#Security>. Accessed: Oct. 24, 2016.
- [65] Microsoft, "Chapter 16: Quality attributes," 2016. [Online]. Available: <http://msdn.microsoft.com/en-us/library/ee658094.aspx#Maintainability>. Accessed: Oct. 24, 2016.
- [66] "Meet Guru99," in *Guru99*. [Online]. Available: http://cdn.guru99.com/images/6-2015/052615_1232_WhatisSDLC01.png. Accessed: Oct. 24, 2016.
- [67] "Annual Report 2014," National Eye Hospital, Colombo, 2014. [Online]. Available: <http://nationaleyehospital.health.gov.lk/downloads/Annual%20Report-2014.pdf>. Accessed: Oct. 24, 2016.
- [68] "Annual Report 2014," National Eye Hospital, Colombo, 2014. [Online]. Available: <http://nationaleyehospital.health.gov.lk/downloads/Annual%20Report-2014.pdf>. Accessed: Oct. 24, 2016.

GLOSSARY

AMD	A medical condition which may result in blurred or no vision in the center of the visual field.
Amsler grid	The Amsler grid is a tool that eye doctors use to detect vision problems resulting from damage to the macula or the optic nerve.
Analog image processing	An analog image processing is any image processing task conducted on two-dimensional analog signals by analog means (as opposed to digital image processing).
Android application	An Android app is a software application running on the Android platform. Because the Android platform is built for mobile devices, a typical Android app is designed for a smartphone or a tablet PC running on the Android OS
Astigmatism	A defect in the eye or in a lens caused by a deviation from spherical curvature
Blepharitis	An inflammation of the eyelid.
Cascade classifiers	First, a classifier (namely a cascade of boosted classifiers working with Haar-like features) is trained with a few hundred sample views of a particular object, called positive examples, which are scaled to the same size, and negative examples - arbitrary images of the same size.
Cataract	A medical condition in which the lens of the eye becomes progressively opaque, resulting in blurred vision.
Chalazion	A cyst in the eyelid that is caused by inflammation of a blocked meibomian gland, usually on the upper eyelid.
Color blindness	Color blindness occurs when you are unable to see colors in a normal way.
Conjunctivitis	An inflammation of the conjunctiva of the eye.
Data extraction	Data extraction involves reducing the amount of resources required

	to describe a large set of data.
Dry eye	An inflammation of the conjunctiva and cornea of the eye, due to inadequate tear secretion.
Edge detection	
Eye lens	The lens is a transparent, biconvex structure in the eye that, along with the cornea, helps to refract light to be focused on the retina.
Glaucoma	A condition of increased pressure within the eyeball, causing gradual loss of sight.
Haar cascade	In the Viola–Jones object detection framework, the Haar-like features are therefore organized in something called a classifier cascade to form a strong learner or classifier. The key advantage of a Haar-like feature over most other features is its calculation speed.
Histogram equalization	Histogram equalization is a technique for adjusting image intensities to enhance contrast.
Image classification	Image classification refers to the task of extracting information classes from a multiband raster image.
Image processing	The analysis and manipulation of a digitized image, especially in order to improve its quality.
Iris	A flat, colored, ring-shaped membrane behind the cornea of the eye, with an adjustable circular opening (pupil) in the center.
Jaeger eye chart	The Jaeger chart is an eye chart used in testing near vision acuity. It is a card on which paragraphs of text are printed, with the text sizes increasing from 0.37mm to 2.5mm. This card is to be held by a patient at a fixed distance from the eye dependent on the J size being read.
Knowledge Base	A store of information or data that is available to draw on.
Lazy eye	An eye with poor vision that is mainly caused by underuse, especially the unused eye in a squint.

Luminas	Luminous intensity is a measure of the wavelength-weighted power emitted by a light source in a particular direction per unit solid angle, based on the luminosity function, a standardized model of the sensitivity of the human eye.
Macular degeneration	Macular degeneration occurs when the small central portion of the retina, known as the macula, deteriorates.
Morphological	Morphological operations apply a structuring element to an input image, creating an output image of the same size.
OpenCv	OpenCv is a library of programming functions mainly aimed at real-time computer vision.
Ophthalmologist	An ophthalmologist is a physician who specializes in the medical and surgical care of the eyes and visual system and in the prevention of eye disease and injury.
Optical image processing	All the light that is emitted by a particular point on an object and passes through the optical system falls on the corresponding point on the light-sensitive medium.
Optometrist	A person who practices optometry.
Pre Processing	Subject (data) to preliminary processing.
Pupil	The pupil is a hole located in the center of the iris of the eye that allows light to strike the retina.
Retina	A layer at the back of the eyeball that contains cells sensitive to light, which trigger nerve impulses that pass via the optic nerve to the brain, where a visual image is formed.
Retinoblastoma	A rare malignant tumor of the retina, affecting young children.
Segmentation	Image segmentation is the process of dividing an image into multiple parts.
Servlets	An applet that runs on a server, typically within Java.
SPARQL queries	SPARQL is an RDF query language, that is, a semantic query

	language for databases, able to retrieve and manipulate data stored in Resource Description Framework (RDF) format.
--	---

APPENDICES

Appendix A: Permission letter from eye hospital

Sri Lankan Institute of Information Technology,
New Kandy Rd,
Malabe 10115,
5th May 2016

Director of National Eye Hospital,
Rev. Buddegama Wimalawansa Thero Mawatha,
Colombo 01000.

Permission to capture images of eye patients

We are currently fourth year undergraduates of Sri Lankan Institute of Information Technology, going to build an android application for the final year research. In our research we are going to analyze four eye illnesses namely Blepharitis, Cataract, Chalazion and Conjunctivitis and come up with an android application where any user can use our "Eye Care" application in order to detect above mentioned eye illnesses in few seconds. We need to collect images of above mentioned eye illness patients in order to build up the software. So please be kind enough to give us your valuable cooperation by giving us to take an image of your face in order to build up efficacious software.

Thank you.

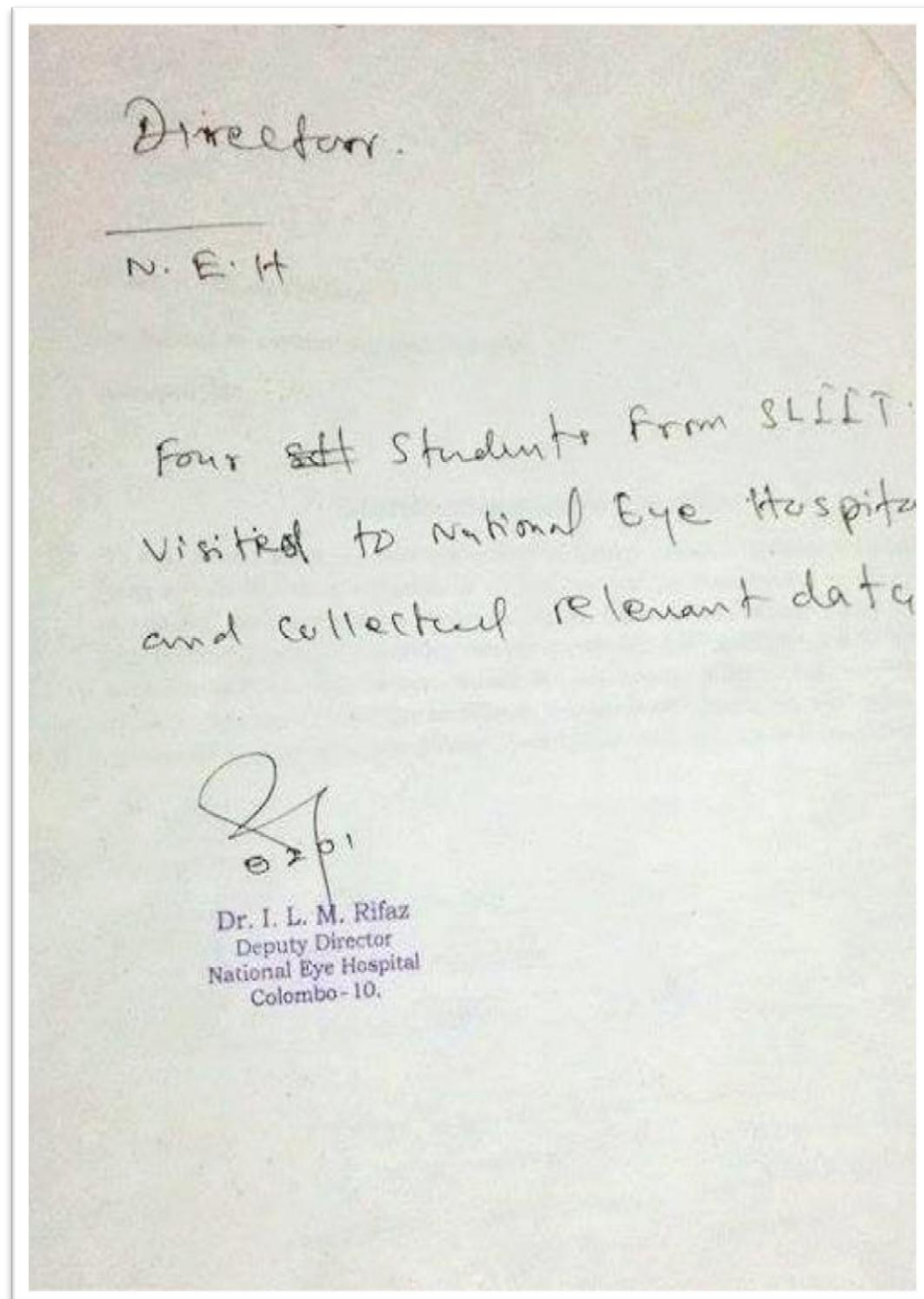
Yours sincerely,

Student Name	Registration Number	Signature
IT13040536	W.A.H Gunarathna	<i>[Signature]</i>
IT13012502	W.M.O.T Wickramasingha	<i>[Signature]</i>
IT13002084	E.A Soysa	<i>[Signature]</i>
IT13006044	M.R.C.I.V Sugathapala	<i>[Signature]</i>

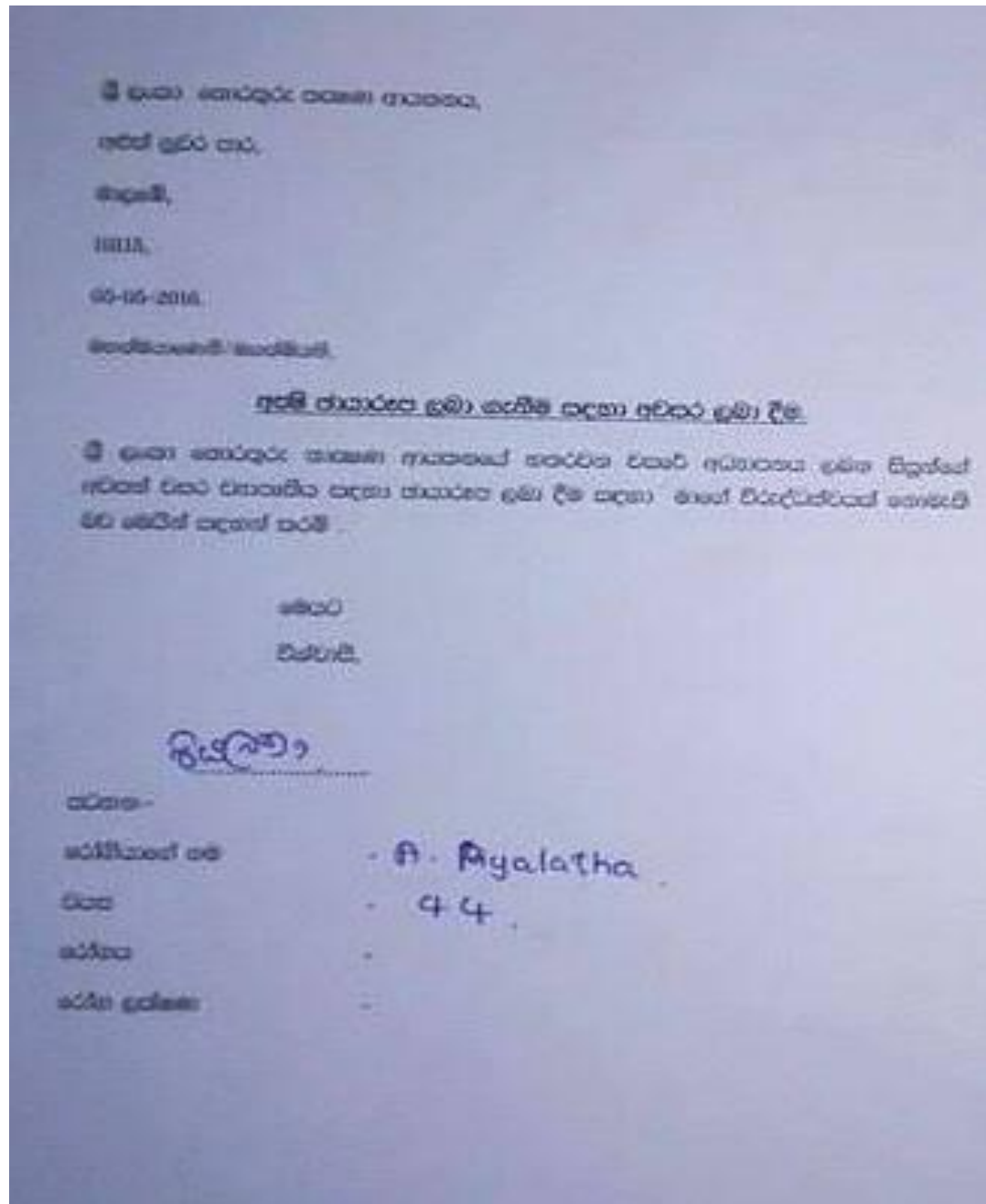
Madam - Please support these students to communicate and take necessary details from two patients without disturbing ward activities, if possible to assist these students.

[Signature]

Appendix B: Letter received from director



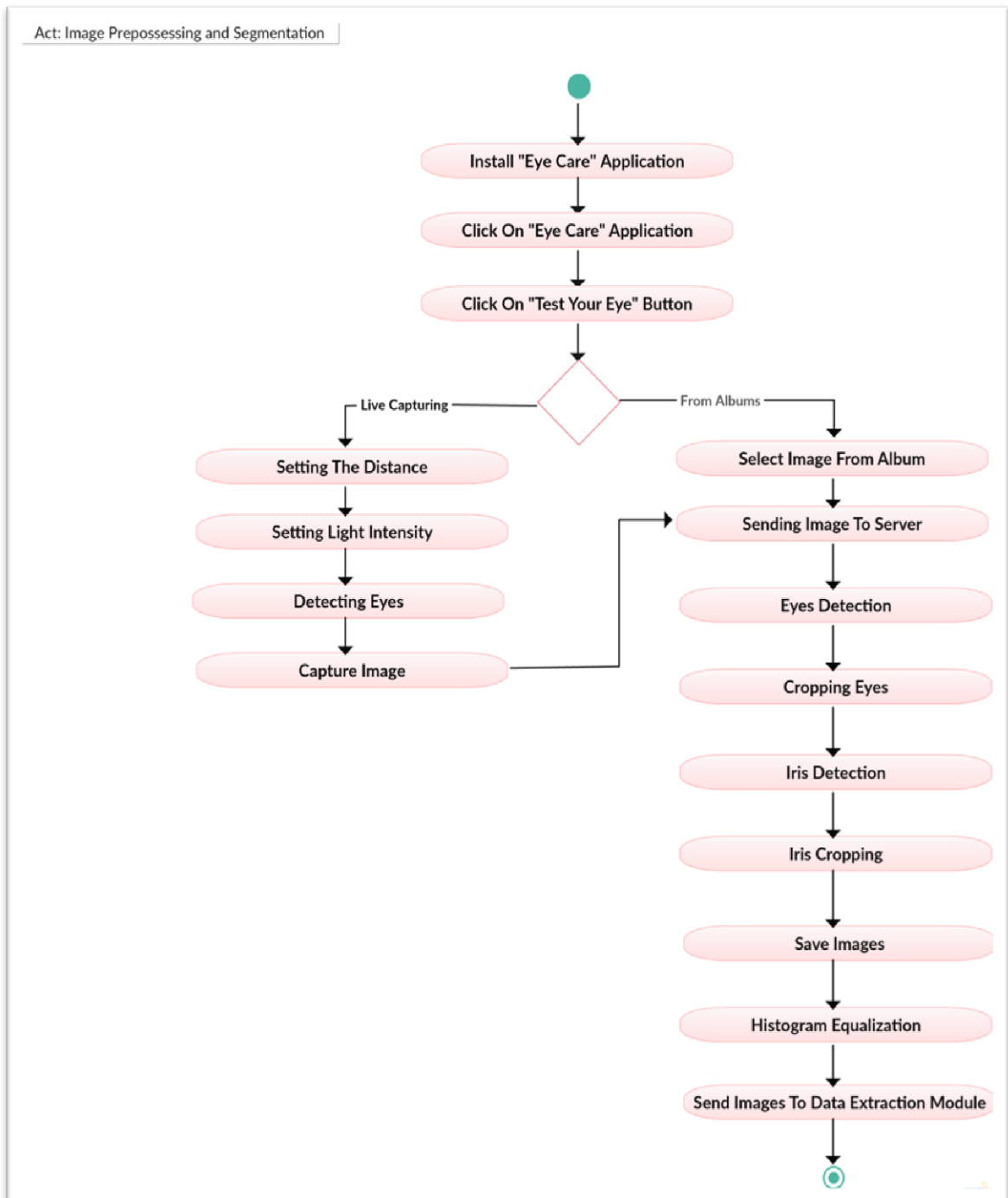
Appendix C: Permission letter from patients



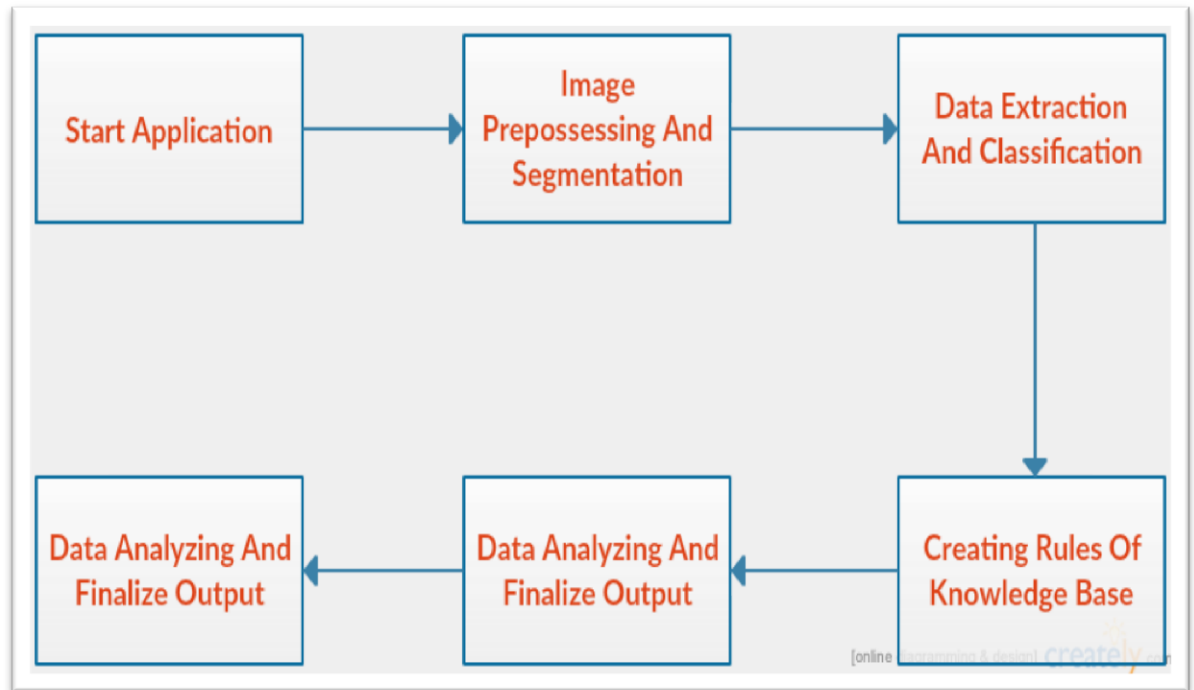
Appendix D: Questions asked by senior consultant

1. What are the common eye illnesses that can find today?
2. How many people are suffering from above eye illnesses?
3. What are the symptoms of the above eye illnesses?
4. What are the unique symptoms of the above illnesses?
5. How are you going to treat the above eye illnesses?
6. What are the home remedies that a patient can take in order to minimize the above illnesses?
7. What are the activities that a patient must not do during the above eye illnesses?
8. Who are the specialists related to the above eye illnesses?
9. How a patient can identifies the above illness at the initial stage?
10. What are the steps to follow when a user suffers from one of the illness mentioned above?

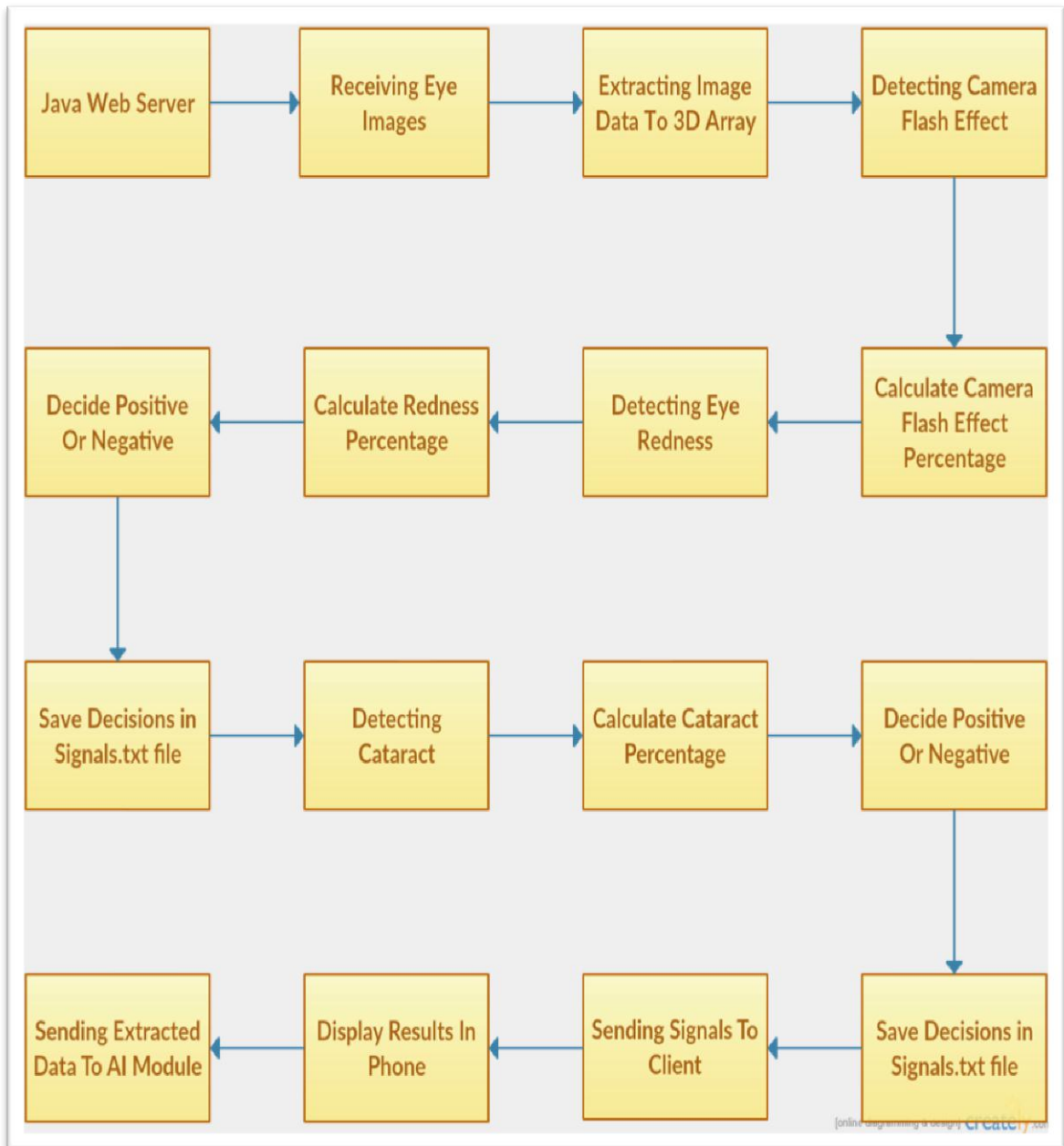
Appendix E: Activity Diagram



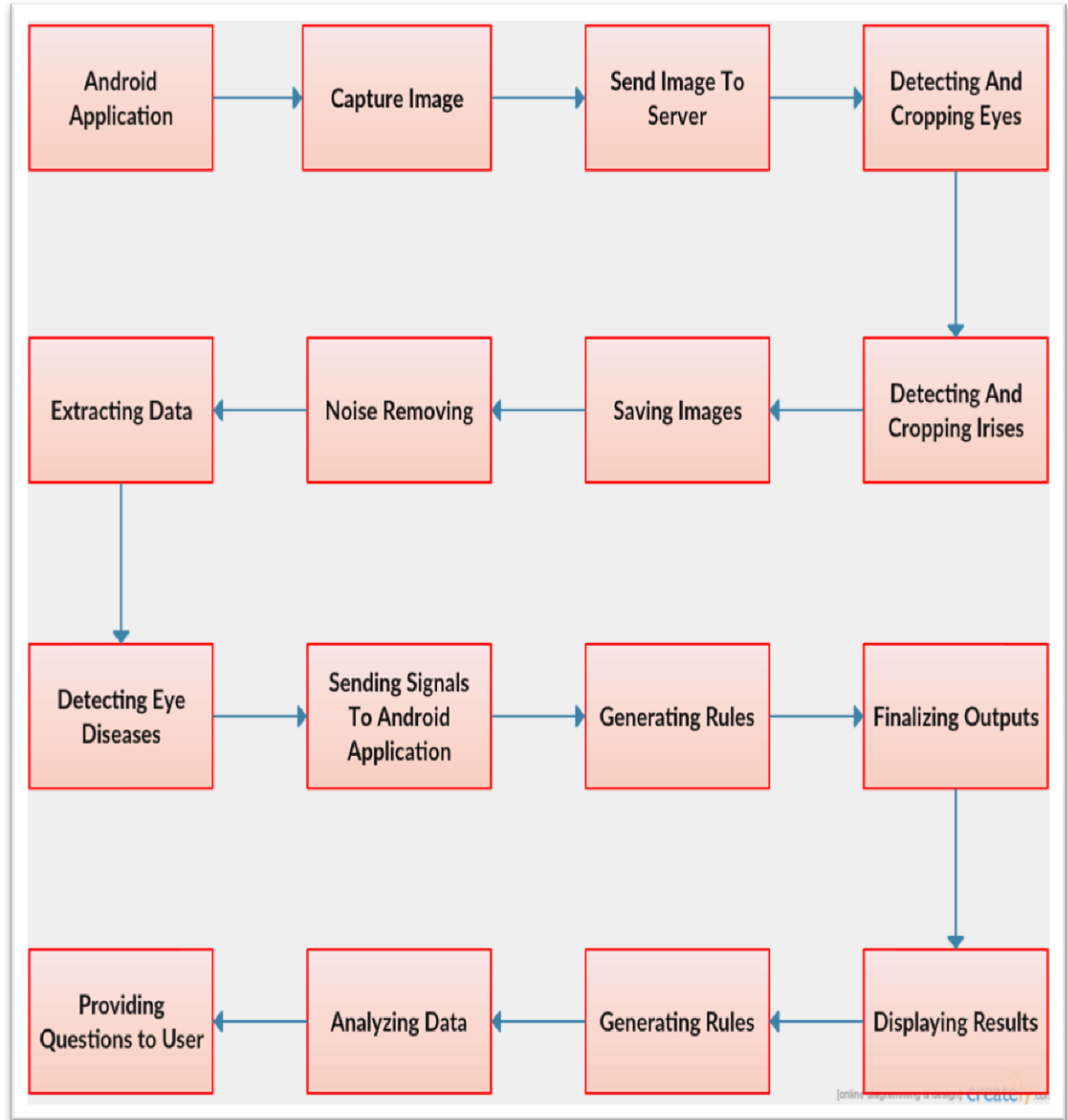
Appendix F: Flow of the modules in Eye Care



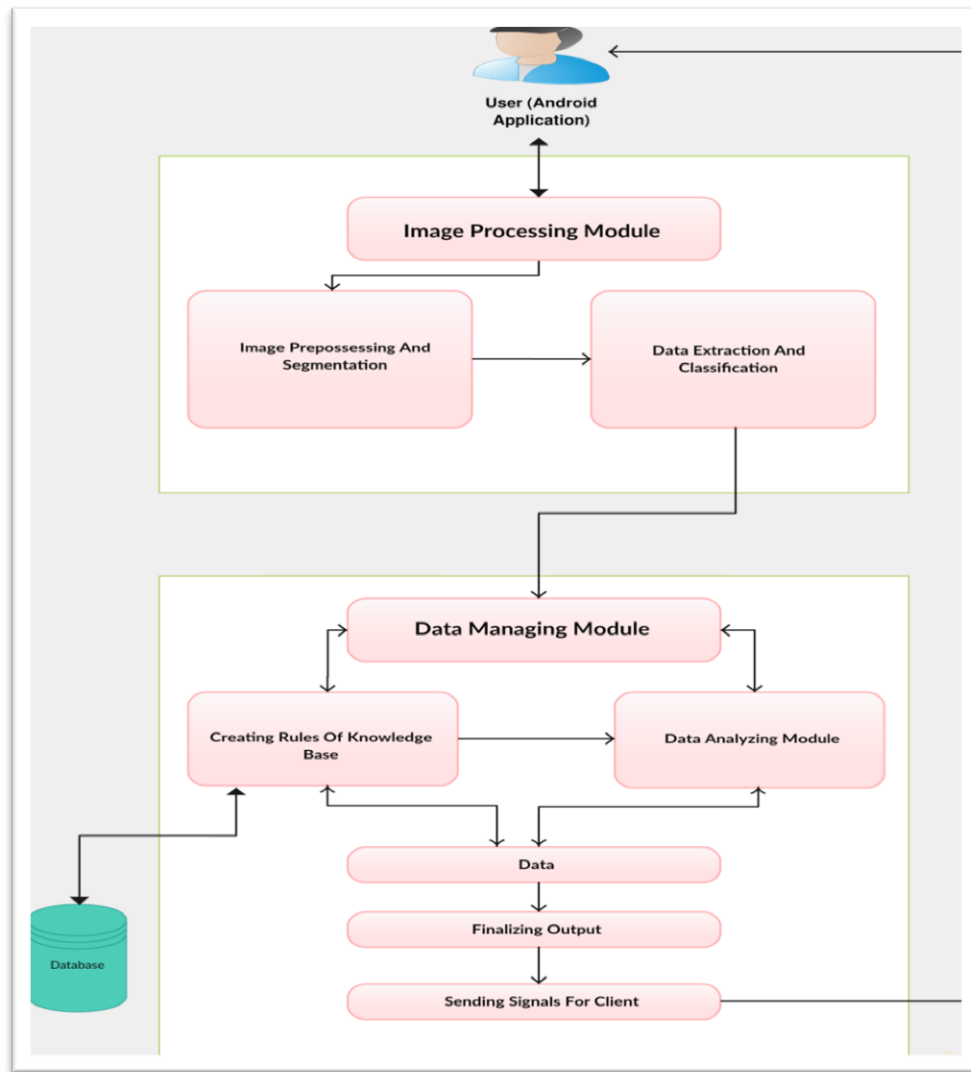
Appendix G: Flow of the system in Eye Care



Appendix H: Flow of image preprocessing and segmentation module



Appendix I: High level architecture diagram



Appendix J: Pseudo code

```
BEGIN

INPUT Image

If Image invalid

THEN

    Display error message

    INPUT image

ELSE

    If Album selection

    THEN

        Select image

        Send image to server

    Endif

    If Live capturing

    THEN

        Capture image

        Send image to server

    Endif

Endif

END
```

Appendix K: Research Paper

Eye Care: Mobile Based Solution For Eye Illness Detection

Umanga Pilapitiya¹, W.M.O.T Wickramasingha², W.A.H Gunarathna³, E.A Soysa⁴,
M.R.C.L.V Sugathapala⁵

Sri Lankan Institute of Information Technology, Malabe, Sri Lanka

Abstract - Eye Care is a mobile based application developed as a solution for problems associated with eye illnesses. People are running out of their time and they do not check their health status daily. As a result of this situation people are suffering from diseases which they even do not know. There is no restriction of using this application by anyone for identifying eye illnesses. Furthermore any person can use this application without relying on a third party. "Eye Care" is the use of Information and Communication Technologies (ICT) for health sector specially to identify the preliminary levels of eye illnesses namely Cataract, Blepharitis, Chalazion and Conjunctivitis. The uniqueness of this research is, it identifies four main eye illnesses which are prominent at present with the use of image processing and semantic web techniques. "Eye Care" identifies above illnesses accurately and efficiently while providing extra knowledge about the illness to the user. Anyone can easily use this application without any challenge as well as a user can operate this android application easily without a single help from a third party.

Key words – Cataract, Conjunctivitis, Blepharitis, Chalazion, Image processing, Semantic Web

I. INTRODUCTION

At present eye illnesses have become prominent due to the negligence of people. People are hesitant to wait in the queues of hospitals since they are busy with their daily schedule. The commonly and largely used solution in this regard is home remedies. Still some home remedies are associated with many disadvantages. When a person takes home remedies for eye illnesses he/she should wait for a longer period of time to cure and since the person does not know the correct dosage can lead to an excessive ingestion. Previous researches in this

field have led to the invention of several mobile based applications. However none of them were provided a complete solution for the eye illnesses. Eye Test app [1], Eye Care Plus app [2], Vision Test 2.0 app [3], EyeXam app [4], Visual acuity Test app [5], White eye detector app [6] and Pink eye [7] app are some of the eye related applications where it is mainly focusing on eye exercise. Pink eye app [7] gives symptoms of Conjunctivitis but not going to detect illnesses. Eye Xam [4] application provides eye exercises but not detecting any eye illness.

In this research paper we present the “Eye Care” mobile based application which detects main four common eye illnesses called Blepharitis [8], Conjunctivitis [9], Chalazion [10] and Cataract [11]. Yet it will overcome the problems of existing mobile based applications and enable the user to use the application by him. The user will no longer have to wait in the queues in the hospitals or try home remedies since Eye Care application facilitates user to check eye illnesses by his/her own.

User can operate the application without getting support from a third party. Eye Care application provides user to find eye consultants and provides immediate home precautions. Since the application does not take any private information of the user, he/she can use this application without any fear. This application enables the user to capture images as well as to choose images from gallery.

To detect eye illnesses the application must identify eyes of the user. Every face has distinguishable landmarks, different peaks and valleys that make up facial features. Human face approximately consisting of 80 nodal points such as width of the nose, depth of eye sockets, shape of cheek bones and the distance between two eyes.

Eye Care mobile based application figures out user's face, left eye and right eye. In order to detect eye illnesses Eye Care application uses semantic web and image processing techniques to get accurate results. This system is designed not only for eye patients. It can use by any user who wish to check eyes.

Illnesses which are detected by eye applications are Conjunctivitis, Cataract, Chalazion and Blepharitis. Conjunctivitis which is also known as pink eye is an inflammation of the conjunctiva. Blepharitis is a common eye condition that causes inflammation of the eye lid. Chalazion is a small bump that appears on the eye lid because of a blocked oil gland. Cataract is a cloud of the eye lens that affects the vision.

Summary of the survey

As in the survey all android based applications address only eye exercises [1]. There couldn't find any application which detects eye illnesses. To overcome those problems Eye Care eye illnesses detecting application is introduced which is designed to identify most prominent four eye illnesses. This application is designed in an attractive way so that patients can interact with the application very easily.

II. METHODOLOGY

The intention of this research project is to function as an eye detecting application. In order to detect eye illnesses image processing and semantic web techniques have used. The entire procedure will be sub divided into four modules as Image preprocessing and segmentation, data extraction and segmentation, Creating rules by knowledge base and Finalizing output by the results gained by data extraction and knowledge base.

- Image preprocessing and segmentation

First Eye Care android application facilitates user to enter an image with the help of camera or else user can choose an image from the gallery which is in the android phone. Eye Care application supports user by displaying the distance in the phone.

Energy of light around the user is measured by this application by using the light sensor available in the phone which avoids user to capture images in darker places. In order to get a high quality image user must need to maintain the distance between user and the android phone. Then image is uploading to the server to detect eye illnesses. Apache tomcat V8.0 is used as a server. First task of the server application is to detect and crop eyes and irises from the image received by android application. In order to detect and crop eyes Haarcascade files were used. There are no Haarcacade files available to detect and crop irises. By using Haar training stuff tool, Eye Care has trained a Haarcasade file to detect and crop irises. Cropped irises and eyes are then sending to extract relevant data to identify eye illnesses separately. Figure 1 show how it detected and cropped eyes by suing Haarcascade files.

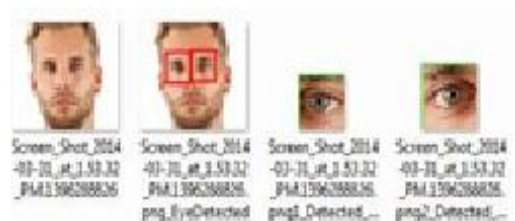


Figure 1: Eye Detection and cropping

- Data Extraction and Classification

Server has chosen as a java dynamic web project. Image which is captured by a user contains several noises. One of them is camera flash effect. Below mentioned figure shows a camera flash effect image.



Figure 2: Camera Flash Effect

In order to remove this camera flash effect noise, each and every pixel information was note down by using a pixel finder tool. Maximum pixel values of the camera flash effect are mentioned below with the aid of a table.

Table 1: Pixel Values

Pixel	Value
Red	230
Green	223
Blue	216

If the pixels are more than the above range it is considered as a camera flash effect pixel. These values are found by using RGB color model.

Conjunctivitis which is the redness of the eye is analyzed by using HSB color model. An equation must be used in order to detect Conjunctivitis.

$$P=H*360 \quad (1)$$

H is the extracted hue pixel and P is the new pixel. If P is between the range of 330 and 360 it can conclude that the pixel is a Conjunctivitis affected pixel.

Cataract which means the white cloud of the iris was identified by using RGB color model. Every single pixel was note down by using pixel founder tool. Below mentioned are the values gained by the pixel founder tool related to Cataract illness.

Table 2: Pixel values of Cataract

Pixel	Value	
	Min	Max
Red	100	170
Green	100	175
Blue	100	195

If the pixel value is between the above min and max range it can conclude that the pixel is a cataract affected pixel. After detecting the illness then the results are sending to knowledge base in order to detect the illness further.

- Creating knowledge Base

The implementation of ontology was made relying on the framework based on knowledge and ontology editor called Protégé. The ontology was encoded using the Web Ontology Language (OWL), which provides classes, properties, individuals and data type values. When creating the knowledge base it uses the data and get from the data extraction process. Here it is training the system dynamically to identify the illnesses. Many applications require the ability to work with well-defined portions of ontology, which are usually called modules. The OWL API contains basic functionality for extracting syntactic-locality based modules [12] from ontologies. These kinds of

modules are based on the notion of conservative-extensions [13], and guarantee to preserve entailments from ontology over a given signature. As an initial step first creates the ontology and then updates it dynamically.

To build the ontology, we gathered data that extracted from image processing. Analyzed hundreds of images to come up with a conclusion that in which range is belong to inflected eye or not. After then we came up with the final ranges and then added those values into ontology. After built the ontology statically then have to update it in a dynamic way. To do it, it must load the existing ontology into our dynamic web project and implemented the code.

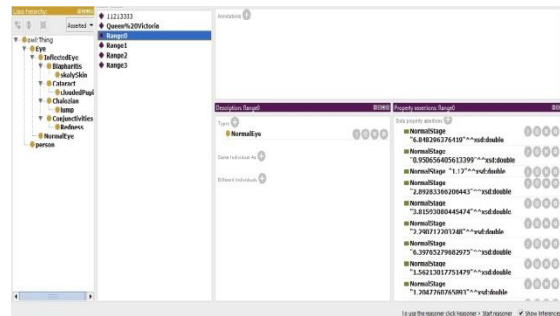


Figure 3: Ontology class structure

Table 3: Ranges of conjunctivitis

Stage of the conjunctivitis	Range (%)
Normal stage	0-7
Initial stage	7.1-10.00
Average stage	10.1-17.5
Critical/Severe stage	Above 17.6

- Finalizing output

In component Data analyzing and finalize output, it matches the input data with the knowledgebase rules and then writes SPARQL select queries to retrieve data. According to the input value which gained by data extraction, here it creates the select queries for the knowledge base to retrieve class name for a given individual value. While this process is handling in the server, android application prompts questions to the user to answer them thoroughly.

Those questionnaires belong to four illness categories name as Conjunctivitis, Chalazion, Blepharitis and Cataract. These questions mainly based on some of the major symptoms of above illnesses. By analyzing answers provided by the user then decides the eye condition of the user.

According to the answers it calculates a percentage value of having the illness. These percentage values categorize in to four stages as Normal stage, Average stage, Critical stage and severe stage.

In server it retrieves the illness name from the knowledge base and by using a servlet it sends the result as a signal to the android application. Finally it gives the eye condition by analyzing answers of the questionnaires and output data retrieve from the knowledge base. And then displays the output to the user. It gives a brief description about the illness and the details about the consultants and hospitals near the user.

III. RESEARCH FINDINGS

There are numerous Haar cascade files to detect eyes, faces and various objects. But still researchers have failed to develop a Haar cascade file to detect iris in java language. In “Eye Care” application it has developed a Haar cascade file for iris detection.

Measuring the light source option helps the client to get a quality image with recommended light energy which uses smart phone light sensor to measure the light energy. If the light energy is lower than the recommended range (1000lu) android phone is not allowing capturing the image.

In distance calibration the distance between user and the android application must be calculated in order to maintain the quality of the image. Here the android phone will guide the user to set the distance correctly. When user zoom the camera, it changes the distance between two eyes. When the distance is satisfied user can capture the image.

IV. DISCUSSION

Eye Care application detects four main eye illnesses called Conjunctivitis, Blepharitis, Cataract and Chalazion by using image processing and semantic web techniques.

First the input image is segmented for irises and eyes and then it extracts relevant data. Extracted data are sending to the knowledge base for further processing and then it finalize the output with the aid of results gained by knowledge base, extracted data and the answers provided by the user.

Following is the accuracy rate of the Eye Care application which was gained by testing 150 images. This is considered as system testing.

No of input images: 150

No of correct results: 125

Accuracy = $(125/150)*100\%$

= 83.33%

According to the above answer it can be concluded that the accuracy of the Eye Care system is about 83.33%.

V. CONCLUSION

In this paper we present the development of the eye detecting application, which we believe is the ideal solution for eye illnesses patients. We proposed an android based mobile application where it identifies most prominent eye illnesses with the use of image processing and semantic web techniques. Eye Care application provides the users with various benefits over existing applications which are available in the market. Those who use this application they can identify the eye illnesses within a short period of time with the aid of android phone and generate results including home precautions that he/she can take to reduce the illness up to some extent and finally they can consult a doctor to cure the illness completely.

There are considerable amount of eye related mobile applications are developed in the field of information technology. Our approach is completely automatic and optimized solution comparing with all others existing. Eye Care application allows user to operate it by his own, without relying on others support. The only limitation of this application is that the Eye Care application detects only four illnesses.

VI. FUTURE WORK

With the use of image processing techniques it detects and crop irises of the provided image. In iris detection and iris cropping it detects and crop irises accurately with some of the garbage values.

Here the system cannot identify the iris directly. When user checks the image folder he/she can view the irises photos clearly. So this is one of the limitations of this application. In order to overcome the above mentioned problems developer can train a Haar cascade file where it needs about 60 hours to create the Haar cascade file. User

must enter about 2000 images of positive images and 5000 images of negative images in order to create a better Haar cascade file.

Here quality of the picture is an issue. If user captures an image from a poor quality camera it is hard to identify iris and eyes in the server application. Therefore in order to minimize the above problem developer must build new Haar cascade files or algorithms which supports for the low quality cameras.

If the image contains a group photo or else other images server is falling down since the application is detecting only two eyes in a human image. If someone wishes to improve Eye Care application he/she can find a way to detect eyes in a group photo. This application is only focusing on four eye illnesses. Therefore a person can improve the system which can detect several illnesses. Eye Care application support only for android phones. Therefore a developer can build Eye Care application which will support for several platforms.

VII. ACKNOELEDGEMENT

Our first honorable gratitude goes to Dr. I. L. M. Rifaz director of National Eye Hospital for giving us information about eye illnesses and allow us to capture photos of patients in order to launch our project. And also we convey our real sense of gratitude and thankfulness to the patients of National Eye Hospital who allowed us to capture images of them.

VIII. REFERENCES

[1] D. Health and Fitness, "Eye test - Android Apps on Google play," in GooglePlay, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.b2creativedesigns.eyetest&hl=en>. [Accessed: Jan. 12, 2016].

- [2] healthcare mobile 4 Health and Fitness, "Eye exercises - eye care plus – Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.eyexamtest.eyecareplus>. [Accessed: Jan. 12, 2016].
- [3] R. L. Health and Fitness, "Vision test 2.0 - Android Apps on Google play," in Google Play, 2015. [Online]. Available: <https://play.google.com/store/apps/details?id=com.threesidedcube.visiondroid&hl=en>. [Accessed: Jan. 12, 2016].
- [4] G. EyeVentures, "EyeXam on the App store," App Store, 2016. [Online]. Available: <https://itunes.apple.com/en/app/eyexam/id357290178?mt=8>. [Accessed: Jan. 12, 2016].
- [5] healthcare mobile 4 Health and Fitness, "Visual Acuity test - Android Apps on Google play," in Google Play, 2016. [Online]. Available: <https://play.google.com/store/apps/details?id=com.eyexamtest.acuity&hl=en>. Accessed: Jan. 12, 2016.
- [6] G. EyeVentures, "EyeXam on the App store," App Store, 2016. [Online]. Available: <https://itunes.apple.com/en/app/eyexam/id357290178?mt=8>. [Accessed: Jan. 22, 2016].
- [7] WebMD, "What is Conjunctivitis, or Pinkeye?," in WebMD, WebMD, 2005. [Online]. Available: <http://www.webmd.com/eye-health/eye-health-conjunctivitis>. [Accessed: Jul. 25, 2016].
- [8] M. MacGill and H. Webberley, "Blepharitis: Causes, symptoms and treatment," Medical News Today, 2016. [Online]. Available: <http://www.Medicalnewstoday.com/articles/185155.php>. [Accessed: Jul. 27, 2016].
- [9] A. A. Azari and N. P. Barney, "Conjunctivitis," Conjunctivitis, vol. 310, no. 16, Oct. 2013. [Online]. Available: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4049531/>. [Accessed: Jul. 27, 2016].

- [10] B. Chou, "Chalazion," in allaboutvision, All About Vision, 2016. [Online]. Available: <http://www.allaboutvision.com/conditions/chalazion.htm>. [Accessed: Jul. 31, 2016].
- [11] G. Bailey, "Hazy, blurry vision means you might have cataracts. Learn more in this complete guide to cataracts," All About Vision. [Online]. Available: <http://www.allaboutvision.com/conditions/cataracts.htm>. Accessed: Jan. 6, 2016.
- [12] Bernardo Cuenca Grau, Ian Horrocks, Yevgeny Kazakov, and Ulrike Sattler. Just the right amount: Extracting modules from ontologies. In WWW 2007, Proceedings of the 16th International World Wide Web Conference, Banff, Canada, May 8-12, 2007, pages 717–727, 2007.
- [13] Silvio Ghilardi, Carsten Lutz, and Frank Wolter. Did I Damage My Ontology? A Case for Conservative Extensions in Description Logic. In Patrick Doherty, John Mylopoulos, and Christopher A. Welty, editors, The 10th International Conference on Principles of Knowledge Representation and Reasoning (KR 2006), Lake District, United Kingdom. AAAI Press, June 2006.